

New Question Bank

Official College Board Digital SAT Question Bank – SAT Math

Algebra → Advanced Math → Problem-Solving & Data Analysis → Geometry & Trigonometry → June Resit

Περιεχόμενα - Contents

1. Algebra	Αλγεβρα · 23 questions.....	2
2. Advanced Math	Προχωρημένα Μαθηματικά · 27 questions	21
3. Problem-Solving & Data Analysis	Ανάλυση Δεδομένων · 19 questions	45
4. Geometry & Trigonometry	Γεωμετρία & Τριγωνομετρία · 51 questions.....	62
5. June Resit 2026 – Math (Part 1)	10 ασκήσεις με αναλυτικές λύσεις	100

Γραμμένα από τον Τσαλαπάτη Σταύρο

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Algebra

Math · 23 questions · Εκφώνηση / Correct Answer / Step-by-Step Solution

Linear equations in one variable

5 questions · Easy 2 / Medium 2 / Hard 1

Linear equations in one variable — Q1 [Easy] (ID: 5eb4918d)

Question ID: 5eb4918d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

If $9x + 4 = 67$, what is the value of $90x + 40$?

Answer

- A. 7
- B. 70
- C. 130
- D. 670

Correct Answer: D

Step-by-Step Solution:

- Step 1: Write down the given equation: $9x + 4 = 67$.
- Step 2: Notice $90x + 40 = 10 \times (9x + 4)$.
- Step 3: Multiply both sides of the original equation by 10: $10(9x + 4) = 10(67)$, which gives $90x + 40 = 670$.
- Step 4: The answer is 670, matching choice D.

Linear equations in one variable — Q2 [Easy] (ID: 935f3403)

Question ID: 935f3403

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

For a restaurant owner to purchase a pizza oven at a total price of 13,000 dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. The equation $13,000 = 2,800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. Which of the following is the best interpretation of 200 in this context?

Answer

- A. The amount, in dollars, of the down payment
- B. The amount, in dollars, of each fixed monthly payment
- C. The total amount, in dollars, paid for the pizza oven after t fixed monthly payments
- D. The total number of fixed monthly payments

Correct Answer: B

Step-by-Step Solution:

- Step 1: Identify what t represents: the number of monthly payments made (given directly in the problem).
- Step 2: Look at the structure of the equation: $13,000 = 2,800 + 200t$. This matches the general pattern (total price) = (down payment) + (payment amount) $\times$ (number of payments).
- Step 3: Since 200 is being multiplied by t , and t counts monthly payments, 200 must be the dollar amount of each individual monthly payment.
- Step 4: Match this interpretation to the choices: 'the amount, in dollars, of each fixed monthly payment' is choice B.

Linear equations in one variable — Q3 [Medium] (ID: 088bc84e)

Question ID: 088bc84e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

If $2(5x) = 6$, what is the value of $5x$?

Answer

- A. $\frac{4}{5}$
- B. $\frac{5}{3}$
- C. 3
- D. 4

Correct Answer: C

Step-by-Step Solution:

- Step 1: Start with the given equation: $2(5x) = 6$.
- Step 2: Divide both sides by 2 to isolate the grouped expression ($5x$): $5x = 6 \div 2 = 3$.
- Step 3: The value of $5x$ is 3, matching choice C.

Linear equations in one variable — Q4 [Medium] [Grid-In] (ID: f8fa0cf0)

Question ID: f8fa0cf0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$$9r = 7(r + 7)$$

What value of r is the solution to the given equation?

Correct Answer: 49/2 or 24.5 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Write the given equation: $9r = 7(r + 7)$.
- Step 2: Distribute the 7 across the parentheses on the right side: $9r = 7r + 49$.
- Step 3: Subtract $7r$ from both sides to collect the r -terms on the left: $9r - 7r = 49$, which simplifies to $2r = 49$.
- Step 4: Divide both sides by 2 to isolate r : $r = 49/2 = 24.5$.
- Step 5: This is a grid-in (student-produced response) question, so you would enter either 49/2 or 24.5 into the answer grid — both are accepted as correct.

Linear equations in one variable — Q5 [Hard] (ID: 55ea0659)

Question ID: 55ea0659

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$x(r - 9) + 4 = 19x + 27$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r ?

Answer

- A. 4
- B. 9
- C. 23
- D. 28

Correct Answer: D

Step-by-Step Solution:

- Step 1: Start with $x(r - 9) + 4 = 19x + 27$.
- Step 2: Distribute x across $(r - 9)$: $xr - 9x + 4 = 19x + 27$.
- Step 3: Move all x -terms to the left side and constants to the right: $xr - 9x - 19x = 27 - 4$.
- Step 4: Combine like terms: $xr - 28x = 23$, which factors to $x(r - 28) = 23$.
- Step 5: For this equation to have exactly one solution, the coefficient $(r - 28)$ must NOT be zero — because if it were zero, the left side would be 0, and $0 = 23$ is impossible (no solution at all, not one).
- Step 6: Set $r - 28 = 0$ to find the forbidden value: $r = 28$. This is the value r cannot be, matching choice D.

Linear equations in two variables

2 questions · Easy 0 / Medium 1 / Hard 1

Linear equations in two variables — Q1 [Medium] (ID: 4722abd2)

Question ID: 4722abd2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

Line p is defined by $2y + 8x = 11$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

Answer

- A. -4
- B. $-\frac{1}{4}$
- C. $\frac{1}{4}$
- D. 4

Correct Answer: C

Step-by-Step Solution:

- Step 1: Start with the given equation for line p : $2y + 8x = 11$.
- Step 2: Subtract $8x$ from both sides: $2y = -8x + 11$.
- Step 3: Divide every term by 2 to fully isolate y : $y = -4x + 11/2$.
- Step 4: Identify the slope of line p from this slope-intercept form: the slope is -4 (the coefficient of x).
- Step 5: To find the slope of a line perpendicular to p , take the negative reciprocal of -4 . Flip the fraction ($-4 = -4/1$ becomes $-1/4$) and then flip the sign: the negative reciprocal is $1/4$.
- Step 6: The slope of line r is $1/4$, matching choice C.

Linear equations in two variables — Q2 [Hard] [Grid-In] (ID: 107d0c62)

Question ID: 107d0c62

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard ■■■■■

Question

$$0.10x + 0.20y = 0.17(x + y)$$

The equation gives a volume x , in gallons, of a 10% solution that could be mixed with a volume y , in gallons, of a 20% solution to produce a 17% solution. According to this equation, what volume, in gallons, of the 20% solution could be mixed with 90.0 gallons of the 10% solution to produce a 17% solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

Correct Answer: 210 (Student-Produced Response — no answer choices given)**Step-by-Step Solution:**

Step 1: Start with the given equation: $0.10x + 0.20y = 0.17(x + y)$.

Step 2: Substitute the known value $x = 90$ into the equation: $0.10(90) + 0.20y = 0.17(90 + y)$.

Step 3: Simplify the left side: $0.10 \times 90 = 9$, so the left side becomes $9 + 0.20y$.

Step 4: Distribute the 0.17 on the right side: $0.17 \times 90 = 15.3$, and $0.17 \times y = 0.17y$, so the right side becomes $15.3 + 0.17y$.

Step 5: Now the equation reads: $9 + 0.20y = 15.3 + 0.17y$.

Step 6: Subtract $0.17y$ from both sides: $9 + 0.03y = 15.3$.

Step 7: Subtract 9 from both sides: $0.03y = 6.3$.

Step 8: Divide both sides by 0.03: $y = 6.3 / 0.03 = 210$.

Step 9: This is a grid-in question — enter 210 as your final answer, representing 210 gallons of the 20% solution.

Systems of two linear equations in two variables

7 questions · Easy 1 / Medium 3 / Hard 3

Systems of two linear equations in two variables — Q1 [Easy] (ID: 0e926898)

Question ID: 0e926898

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question
Students and chaperones from a middle school are going on a field trip to a museum. Admission to the museum will cost \$8.50 for each student and \$12 for each chaperone. It will cost a total of \$730 for admission to the museum for all 83 people on the field trip. How many students are going on the field trip?

- Answer**
- A. 31
 - B. 70
 - C. 76
 - D. 86

Correct Answer: C

Step-by-Step Solution:

Step 1: Define variables: let s = number of students, c = number of chaperones.

Step 2: Write the total-people equation: $s + c = 83$.

Step 3: Write the total-cost equation: $8.50s + 12c = 730$.

Step 4: Solve the first equation for c : $c = 83 - s$.

Step 5: Substitute this expression for c into the second equation: $8.50s + 12(83 - s) = 730$.

Step 6: Distribute the 12: $8.50s + 996 - 12s = 730$.

Step 7: Combine the s -terms: $8.50s - 12s = -3.5s$, so the equation becomes $-3.5s + 996 = 730$.

Step 8: Subtract 996 from both sides: $-3.5s = -266$.

Step 9: Divide both sides by -3.5 : $s = 76$.

Step 10: This matches choice C — 76 students are going on the field trip.

Systems of two linear equations in two variables — Q2 [Medium] (ID: ea8bd66f)

Question ID: ea8bd66f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y = 4x$$
$$5x - 3y = -14$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

Answer

- A. 4
- B. 6
- C. 8
- D. 10

Correct Answer: D

Step-by-Step Solution:

- Step 1: Note that the first equation is already solved for y: $y = 4x$.
- Step 2: Substitute $4x$ in place of y into the second equation: $5x - 3(4x) = -14$.
- Step 3: Distribute the -3 : $5x - 12x = -14$.
- Step 4: Combine like terms: $-7x = -14$.
- Step 5: Divide both sides by -7 : $x = 2$.
- Step 6: Substitute $x = 2$ back into $y = 4x$ to find y : $y = 4(2) = 8$.
- Step 7: The question asks for $x + y$, so calculate: $2 + 8 = 10$.
- Step 8: This matches choice D.

Systems of two linear equations in two variables — Q3 [Medium] (ID: 340afb12)

Question ID: 340afb12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

Ellen bought two types of snack bags for a school event. Each box of chips Ellen bought contained 24 bags and cost \$11. Each box of crackers Ellen bought contained 16 bags and cost \$9. It cost Ellen \$497 before tax to buy 968 bags. If x is the number of boxes of chips and y is the number of boxes of crackers that Ellen bought, which of the following systems of equations represents this situation?

Answer

- A. $11x + 16y = 968$
 $24x + 9y = 497$
- B. $11x + 9y = 968$
 $24x + 16y = 497$
- C. $11x + 9y = 497$
 $24x + 16y = 968$
- D. $9x + 11y = 497$
 $16x + 24y = 968$

Correct Answer: C

Step-by-Step Solution:

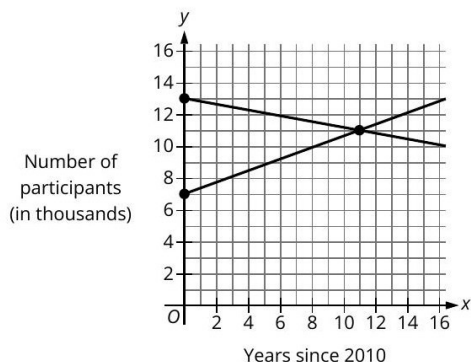
- Step 1: Identify what x and y represent: x = number of boxes of chips, y = number of boxes of crackers.
- Step 2: Build the COST equation using the price per box: chips cost \$11 per box, crackers cost \$9 per box, and the total cost was \$497. This gives: $11x + 9y = 497$.
- Step 3: Build the BAGS equation using the bags per box: chips have 24 bags per box, crackers have 16 bags per box, and the total was 968 bags. This gives: $24x + 16y = 968$.
- Step 4: Compare these two equations to the answer choices — choice C lists exactly $11x + 9y = 497$ and $24x + 16y = 968$, matching perfectly.

Systems of two linear equations in two variables — Q4 [Medium] (ID: 080e75c2)

Question ID: 080e75c2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question



The functions f and g model the number of participants, in thousands, in two different programs x years since 2010. The graphs of $y = f(x)$ and $y = g(x)$ are shown. Which of the following could represent functions f and g ?

Answer

A. $f(x) = -\frac{4}{11}x + 13$

$g(x) = \frac{2}{11}x + 7$

B. $f(x) = -\frac{4}{11}x + 7$

$g(x) = \frac{2}{11}x + 13$

C. $f(x) = -\frac{2}{11}x + 13$

$g(x) = \frac{4}{11}x + 7$

D. $f(x) = -\frac{2}{11}x + 7$

$g(x) = \frac{4}{11}x + 13$

Correct Answer: C

Step-by-Step Solution:

Step 1: Identify the y-intercepts directly from the graph: the decreasing line crosses the y-axis at 13, and the increasing line crosses the y-axis at 7.

Step 2: Identify a second clear point on each line — both lines appear to cross paths at approximately (11, 11).

Step 3: Calculate the slope of the decreasing line using the two points (0, 13) and (11, 11): slope = $(11 - 13) / (11 - 0) = -2/11$.

Step 4: Calculate the slope of the increasing line using the two points (0, 7) and (11, 11): slope = $(11 - 7) / (11 - 0) = 4/11$.

Step 5: Write each equation in slope-intercept form: $f(x) = -2/11x + 13$ (the decreasing line) and $g(x) = 4/11x + 7$ (the increasing line).

Step 6: Compare to the answer choices — choice C lists $f(x) = -2/11x + 13$ and $g(x) = 4/11x + 7$, matching exactly.

Systems of two linear equations in two variables — Q5 [Hard] (ID: 59813abf)

Question ID: 59813abf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$38(x - n) = 38y + 38n$$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

Answer

- A. $4x - 4y = 8n$
- B. $4x + 4y = 4n$
- C. $4x + 4y = 8n$
- D. $4x - 4y = 4n$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Simplify the given equation: $38(x - n) = 38y + 38n$. Divide every term by 38: $x - n = y + n$.
- Step 2: Rearrange into standard form: $x - y = n + n = 2n$.
- Step 3: For a system to have NO solution, the second equation needs the exact same coefficient ratio for x and y (so the lines are parallel) but a DIFFERENT constant value relative to that same scaling (so the lines never actually meet).
- Step 4: Test each answer choice by simplifying it to the same ' $x - y = \dots$ ' form. Choice D: $4x - 4y = 4n$. Divide by 4: $x - y = n$.
- Step 5: Compare: the original equation says $x - y = 2n$, and choice D says $x - y = n$. Since n is a positive constant, $2n$ and n are always different values — confirming these are parallel, non-identical lines with NO solution.
- Step 6: Double-check the other choices don't also produce this pattern — only choice D fits.

Systems of two linear equations in two variables — Q6 [Hard] (ID: 5e9b6079)

Question ID: 5e9b6079

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$y = 7x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

Answer

- A. $-21x + y = 54$
- B. $-21x + y = 18$
- C. $-7x + y = 21$
- D. $-7x + y = 18$

Correct Answer: C

Step-by-Step Solution:

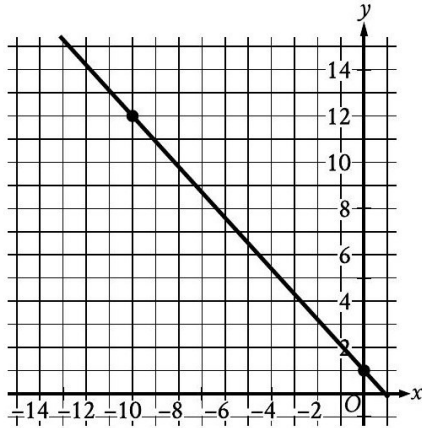
- Step 1: Rewrite the given equation $y = 7x + 18$ into standard form by subtracting $7x$ from both sides: $-7x + y = 18$.
- Step 2: For no solution, look for an answer choice with the identical coefficients on x and y (-7 and 1) but a constant term different from 18 .
- Step 3: Check choice C: $-7x + y = 21$. The coefficients (-7 and 1) match exactly, and the constant (21) is different from 18 — this is the parallel-line pattern needed for no solution.
- Step 4: Confirm this is correct and check the other choices don't also fit this pattern .

Systems of two linear equations in two variables — Q7 [Hard] [Grid-In] (ID: 18ac8354)

Question ID: 18ac8354

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question



The graph of line g is shown in the xy -plane. Line k is defined by $165x + py = w$, where p and w are constants. If line k is graphed in this xy -plane, resulting in the graph of a system of two linear equations, the system of two linear equations will have infinitely many solutions. What is the value of $p + w$?

Correct Answer: 495 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Read two clear points off the graph of line g : it appears to pass through $(0, 2)$ — the y -intercept — and $(-10, 12)$.
- Step 2: Calculate the slope using these two points: $\text{slope} = (12 - 2) / (-10 - 0) = 10 / (-10) = -1$.
- Step 3: Write the equation of line g in slope-intercept form using the y -intercept (2) and slope (-1) : $y = -x + 2$.
- Step 4: Rearrange this into standard form to match the format of line k : $x + y = 2$.
- Step 5: For line k ($165x + py = w$) to represent the exact same line as g ($x + y = 2$) — which is required for infinitely many solutions — every term must be scaled by the same factor. Since the x -coefficient goes from 1 to 165 , the scale factor is 165 .
- Step 6: Apply this scale factor to the other terms: the y -coefficient becomes $1 \times 165 = 165$, so $p = 165$. The constant becomes $2 \times 165 = 330$, so $w = 330$.
- Step 7: Calculate $p + w = 165 + 330 = 495$.
- Step 8: This is a grid-in question — enter 495 as your final answer.

Linear functions

6 questions · Easy 1 / Medium 2 / Hard 3

Linear functions — Q1 [Easy] (ID: 0c490cd5)

Question ID: 0c490cd5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

Sam has a flag with an area of 110 square inches (in^2). Sam plans to enlarge the flag by sewing on pieces of colored nylon, where each piece would increase the flag's area by 64 in^2 . Which equation gives the flag's total area y , in square inches, after Sam sews on x pieces of nylon?

Answer

- A. $y = 64x$
- B. $y = 64 + 110x$
- C. $y = 110 + 64x$
- D. $y = 174x$

Correct Answer: C

Step-by-Step Solution:

- Step 1: Identify the fixed starting value: the flag begins with an area of 110 square inches before Sam adds any nylon pieces.
- Step 2: Identify the rate of change: each nylon piece adds 64 square inches to the area.
- Step 3: Identify the variable: x represents the number of nylon pieces added.
- Step 4: Combine these into the equation: total area = starting area + (rate)(number of pieces), which gives $y = 110 + 64x$.
- Step 5: This matches choice C.

Linear functions — Q2 [Medium] (ID: fc4a0d2a)

Question ID: fc4a0d2a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$g(x) = 3(14x - 15)$$

What is the y -coordinate of the y -intercept of the graph of $y = g(x) - 2$ in the xy -plane?

Answer

- A. -47
- B. -45
- C. -17
- D. -15

Correct Answer: A

Step-by-Step Solution:

- Step 1: Expand the given function: $g(x) = 3(14x - 15) = 3(14x) - 3(15) = 42x - 45$.
- Step 2: Find the y -intercept of $g(x)$ itself by evaluating $g(0)$: $g(0) = 42(0) - 45 = -45$.
- Step 3: The question asks about the NEW function $y = g(x) - 2$, which shifts every output of g down by 2.
- Step 4: Apply this shift to the y -intercept you just found: $-45 - 2 = -47$.
- Step 5: This matches choice A.

Linear functions — Q3 [Medium] (ID: 2493c767)

Question ID: 2493c767

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$s(x) = 133 - 4x$$

The function s shown gives the number of spoons, $s(x)$, remaining in a dispenser x minutes after the dispenser had been loaded with spoons. Which statement is the best interpretation of $s(3) = 121$?

Answer

- A. The number of spoons remaining in the dispenser decreased by a total of 3 spoons after 121 minutes.
- B. There were 3 spoons remaining in the dispenser 121 minutes after the dispenser had been loaded with spoons.
- C. There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons.
- D. The number of spoons remaining in the dispenser decreased by a total of 121 spoons after 3 minutes.

Correct Answer: C

Step-by-Step Solution:

- Step 1: Identify what the input variable x represents in this context: minutes after the dispenser was loaded with spoons.
- Step 2: Identify what the output $s(x)$ represents: the number of spoons remaining in the dispenser at that time.
- Step 3: Interpret $s(3) = 121$ by substituting these roles: the INPUT is 3 (meaning 3 minutes have passed), and the OUTPUT is 121 (meaning 121 spoons remain at that time).
- Step 4: Match this interpretation to the answer choices: 'There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons' is choice C.

Linear functions — Q4 [Hard] (ID: d4572f55)

Question ID: d4572f55

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

x	y
-34	t
-17	$t + 23$
0	$t + 46$

For a linear relationship between x and y , the table gives three values of x and their corresponding values of y , where t is a constant. Which equation represents this relationship?

Answer

- A. $y = -2x + t + 23$
- B. $y = 2x + t + 23$
- C. $y = -\frac{23}{17}x + t + 46$
- D. $y = \frac{23}{17}x + t + 46$

Correct Answer: D

Step-by-Step Solution:

Step 1: Choose two rows from the table where subtracting the y -values will cancel out the unknown t . Use $x = -17$ ($y = t + 23$) and $x = 0$ ($y = t + 46$).

Step 2: Calculate the change in y : $(t + 46) - (t + 23) = t + 46 - t - 23 = 23$ (the t cancels out completely).

Step 3: Calculate the change in x : $0 - (-17) = 17$.

Step 4: Calculate the slope: $(\text{change in } y) / (\text{change in } x) = 23/17$.

Step 5: Since $x = 0$ gives $y = t + 46$ directly from the table, this row already tells you the y -intercept of the line: $t + 46$.

Step 6: Write the full equation in slope-intercept form: $y = (23/17)x + t + 46$.

Step 7: This matches choice D. As a check, verify with the third row ($x = -34$, $y = t$): plug $x = -34$ into the equation: $y = (23/17)(-34) + t + 46 = 23(-2) + t + 46 = -46 + t + 46 = t$. This matches the table exactly, confirming the equation is correct.

Linear functions — Q5 [Hard] (ID: 1a2a0f59)

Question ID: 1a2a0f59

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

A yoga studio is offering a deal where the first session is free, the second session is half off the regular price, and the remaining sessions are at the regular price. If the regular price of a session is \$25.40, which function f gives the total cost, in dollars, of purchasing x sessions using this deal, where $x \geq 2$?

Answer

- A. $f(x) = 25.40(x - 1) + 12.70$
- B. $f(x) = 25.40(x - 2) + 12.70$
- C. $f(x) = 25.40(x - 1) + 12.70(x - 2)$
- D. $f(x) = 25.40(x - 2) + 12.70(x - 1)$

Correct Answer: B

Step-by-Step Solution:

- Step 1: Identify the special pricing for the first two sessions: session 1 is free (\$0), and session 2 is half price ($\$25.40 \div 2 = \12.70).
- Step 2: Add these two special amounts together as a fixed constant: $0 + 12.70 = 12.70$. This fixed amount doesn't change no matter how many total sessions (x) are purchased.
- Step 3: Determine how many sessions are charged at the FULL regular price: since sessions 1 and 2 are special, all sessions from the 3rd onward are regular price — that's $(x - 2)$ sessions when x is the total number of sessions and $x \geq 2$.
- Step 4: Build the full cost function: (fixed special-session cost) + (regular price) $\times$ (number of regular-price sessions) = $12.70 + 25.40(x - 2)$.
- Step 5: This matches choice B: $f(x) = 25.40(x - 2) + 12.70$.

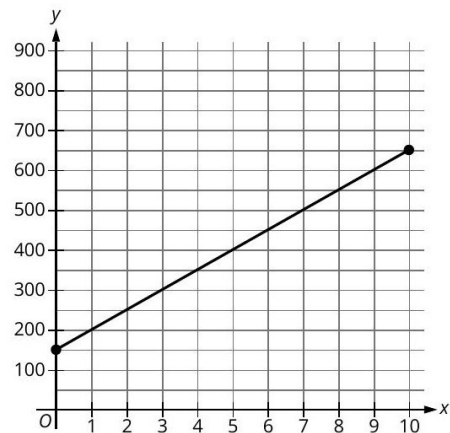
Linear functions — Q6 [Hard] [Grid-In] (ID: 590d662d)

Question ID: 590d662d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance y , in meters, Mary had walked on the trail x minutes after leaving the trail sign.



What distance, in meters, did Mary walk on the trail in the 10 minutes after leaving the trail sign?

Correct Answer: 500 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Read the starting point directly from the graph: at $x = 0$ minutes (right when Mary left the sign), the graph shows $y = 150$ meters already walked.
- Step 2: Read the ending point from the graph: at $x = 10$ minutes, the graph shows $y = 650$ meters total walked.
- Step 3: Since the question asks for the distance walked DURING the 10-minute interval (not the total distance at the 10-minute mark), calculate the CHANGE in y : $650 - 150 = 500$.
- Step 4: This is a grid-in question — enter 500 as your final answer, representing 500 meters walked during those 10 minutes.

Linear inequalities in one or two variables

3 questions · Easy 0 / Medium 3 / Hard 0

Linear inequalities in one or two variables — Q1 [Medium] (ID: 30b1e186)

Question ID: 30b1e186

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

In the xy -plane, which of the following does NOT contain any points that are part of the solution set to $5x - 7y > 35$?

Answer

- A. The x -axis
- B. The region where $x > 0$ and $y > 0$
- C. The region where $x < 0$ and $y < 0$
- D. The region where $x < 0$ and $y > 0$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Start with the given inequality: $5x - 7y > 35$.
- Step 2: Isolate y by first subtracting $5x$ from both sides: $-7y > 35 - 5x$, or equivalently $-7y > -5x + 35$.
- Step 3: Divide both sides by -7 . Remember: dividing an inequality by a **NEGATIVE** number flips the inequality sign. This gives: $y < (5/7)x - 5$.
- Step 4: This tells you the solution set is every point **BELOW** the line $y = (5/7)x - 5$.
- Step 5: Test choice D's region ($x < 0$ and $y > 0$): for any negative x -value, the boundary line $y = (5/7)x - 5$ is already less than -5 (since $(5/7)$ times a negative number is negative, minus 5 makes it even more negative). For a point to satisfy the inequality, y would need to be even smaller (more negative) than this already-negative boundary value.
- Step 6: But choice D requires $y > 0$ (positive), which can never be less than a negative boundary value — so this region contains zero solution points, confirming choice D is correct.

Linear inequalities in one or two variables — Q2 [Medium] (ID: d914abec)

Question ID: d914abec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

To complete a landscaping project, Adam charges a fee of \$20.00 for his equipment and \$9.50 per hour spent working on the project. To complete this same landscaping project, Caroline charges a fee of \$17.00 for her equipment and \$10.00 per hour spent working on the project. If x represents the number of hours spent working on the landscaping project, what are all the values of x for which Caroline's total charge is greater than Adam's total charge?

Answer

- A. $5 \leq x \leq 6$
- B. $6 \leq x \leq 7$
- C. $x < 5$
- D. $x > 6$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Write Caroline's total charge as an expression: her fee is \$17, plus \$10 for every hour (x), giving $17 + 10x$.
- Step 2: Write Adam's total charge as an expression: his fee is \$20, plus \$9.50 for every hour (x), giving $20 + 9.5x$.
- Step 3: Set up the inequality for when Caroline's charge is GREATER than Adam's: $17 + 10x > 20 + 9.5x$.
- Step 4: Subtract $9.5x$ from both sides to gather the x -terms on one side: $17 + 0.5x > 20$.
- Step 5: Subtract 17 from both sides: $0.5x > 3$.
- Step 6: Divide both sides by 0.5: $x > 6$.
- Step 7: This matches choice D.

Linear inequalities in one or two variables — Q3 [Medium] (ID: e8fd6d99)

Question ID: e8fd6d99

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

Terry has a goal to collect at least 16 items per week for a donation drive. This week, Terry has collected 4 items. If x represents the additional number of items Terry needs to collect this week to meet the goal, which inequality represents this situation?

Answer

- A. $4 - x \leq 16$
- B. $4 + x \leq 16$
- C. $4 - x \geq 16$
- D. $4 + x \geq 16$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Identify the two quantities being added together: the 4 items Terry has already collected, and x , the additional items still needed.
- Step 2: Write an expression for the total number of items Terry will have collected by the end of the week: $4 + x$.
- Step 3: Translate the phrase 'at least 16 items' into an inequality symbol: 'at least' means the total must be greater than OR EQUAL TO 16, so use the $\geq$ symbol.
- Step 4: Combine these pieces into the final inequality: $4 + x \geq 16$.
- Step 5: This matches choice D.

Advanced Math

Math · 27 questions · Εκφώνηση / Correct Answer / Step-by-Step Solution

Nonlinear functions

11 questions · Easy 2 / Medium 2 / Hard 7

Nonlinear functions — Q1 [Easy] (ID: 3cf2698e)

Question ID: 3cf2698e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The number of cells of a certain bacteria doubles every 25 minutes during its growth period when placed in a medium of milk in a controlled laboratory environment. If a sample of 830 cells of this bacteria is placed in milk in a controlled laboratory environment, which function f best represents the number of cells of the bacteria x minutes after the growth period begins, where x is less than the number of minutes in the growth period?

Answer

- A. $f(x) = 830\left(\frac{1}{2}\right)^{\frac{x}{25}}$
- B. $f(x) = 830(2)^{\frac{x}{25}}$
- C. $f(x) = 830\left(\frac{1}{3}\right)^{\frac{x}{25}}$
- D. $f(x) = 830(3)^{\frac{x}{25}}$

Correct Answer: B

Step-by-Step Solution:

Step 1: Identify the initial amount: 830 cells (given directly in the problem).

Step 2: Identify the growth factor: the population 'doubles,' so the base of the exponential is 2 (not $\frac{1}{2}$ — that would represent something halving, or decaying, not growing).

Step 3: Identify the time period: doubling happens 'every 25 minutes,' so the exponent must be $x/25$ — this way, when $x = 25$ minutes have passed, the exponent is $25/25 = 1$, giving exactly one full doubling.

Step 4: Assemble the full function: $f(x) = (\text{initial amount}) \times (\text{growth factor})^{(x / \text{period})} = 830(2)^{(x/25)}$.

Step 5: This matches choice B.

Nonlinear functions — Q2 [Easy] (ID: fe62f031)

Question ID: fe62f031

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function f is defined by $f(x) = 3x^5$. In the xy -plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. Which equation defines the function g ?

Answer

- A. $g(x) = 3x^5 + 4$
- B. $g(x) = 3x^5 - 4$
- C. $g(x) = 3x^{20}$
- D. $g(x) = 12x^5$

Correct Answer: A

Step-by-Step Solution:

- Step 1: Write down the original function: $f(x) = 3x^5$.
- Step 2: Recall the rule for vertical shifts: shifting a graph UP by k units means adding k to the entire function, giving $g(x) = f(x) + k$.
- Step 3: Substitute $k = 4$ (since the graph shifts up 4 units) and the original function: $g(x) = 3x^5 + 4$.
- Step 4: This matches choice A.

Nonlinear functions — Q3 [Medium] (ID: 71b74a44)

Question ID: 71b74a44

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$y = 7,400(0.87)^x$$

The given equation estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the piece of equipment at the time it was purchased?

Answer

- A. 740
- B. 6,438
- C. 7,400
- D. 8,700

Correct Answer: C

Step-by-Step Solution:

- Step 1: Recognize that 'at the time it was purchased' means $x = 0$ (no years have passed yet).
- Step 2: Substitute $x = 0$ into the given equation: $y = 7,400(0.87)^0$.
- Step 3: Recall that any nonzero base raised to the power of 0 equals 1: $(0.87)^0 = 1$.
- Step 4: Simplify: $y = 7,400 \times 1 = 7,400$.
- Step 5: This matches choice C.

Question ID: 769b612d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The function g is defined by $g(x) = 10^{5x-8}$. What is the value of x when $g(x)$ is equal to 10,000?

Answer

- A. $\frac{11}{5}$
- B. $\frac{12}{5}$
- C. 3
- D. 4

Correct Answer: B

Step-by-Step Solution:

- Step 1: Start with the given equation: $g(x) = 10^{(5x-8)}$, and we want $g(x) = 10,000$.
- Step 2: Set up the equation: $10^{(5x-8)} = 10,000$.
- Step 3: Rewrite 10,000 as a power of 10: $10,000 = 10 \times 10 \times 10 \times 10 = 10^4$.
- Step 4: Now both sides have the same base (10): $10^{(5x-8)} = 10^4$.
- Step 5: Since the bases match, set the exponents equal to each other: $5x - 8 = 4$.
- Step 6: Add 8 to both sides: $5x = 12$.
- Step 7: Divide both sides by 5: $x = 12/5$.
- Step 8: This matches choice B.

Nonlinear functions — Q5 [Hard] (ID: f01108a8)

Question ID: f01108a8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$y = 9\left(\frac{a}{7}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy -plane cross the x -axis, where a , b , and c are positive constants such that $a > 7$ and $b > c$?

Answer

- A. Zero
- B. One
- C. Two
- D. Three

Correct Answer: B

Step-by-Step Solution:

- Step 1: Identify the base of the exponential term: $a/7$. Since the problem states $a > 7$, this means $a/7 > 1$, so the exponential term is a GROWING (increasing) exponential, not a decaying one.
- Step 2: Determine the horizontal asymptote: as $x \rightarrow -\infty$, $(a/7)^{(x+c)} \rightarrow 0$, so $y \rightarrow 0 - b = -b$.
- Step 3: Since b is stated to be a positive constant, $-b$ is a negative number — meaning the graph's horizontal asymptote is below the x -axis (in negative y territory).
- Step 4: Since the function is strictly increasing (from Step 1) and approaches a NEGATIVE value as $x \rightarrow -\infty$ but grows without bound as $x \rightarrow +\infty$, the graph must cross from below the x -axis to above it — and because it's strictly increasing (never doubles back), this crossing happens exactly once.
- Step 5: The condition $b > c$ given in the problem is extra information not needed to answer this particular question — it doesn't change the crossing count, only potentially the crossing's exact location.
- Step 6: This matches choice B: One.

Nonlinear functions — Q6 [Hard] (ID: 58b4c6f3)

Question ID: 58b4c6f3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(2, 13)$. What is the value of ab ?

Answer

- A. 13
- B. 18
- C. 20
- D. 26

Correct Answer: B

Step-by-Step Solution:

- Step 1: Write down the general function: $h(x) = a^x + b$.
- Step 2: Substitute the first point $(0, 10)$: $h(0) = a^0 + b$. Since any nonzero number to the power of 0 equals 1, this simplifies to $1 + b = 10$.
- Step 3: Solve for b : $b = 10 - 1 = 9$.
- Step 4: Substitute the second point $(2, 13)$: $h(2) = a^2 + b = a^2 + 9 = 13$.
- Step 5: Solve for a^2 : $a^2 = 13 - 9 = 4$.
- Step 6: Take the square root of both sides: $a = 2$ (using the positive root, since the problem states a is a positive constant).
- Step 7: Calculate the requested value: $ab = 2 \times 9 = 18$.
- Step 8: This matches choice B.

Nonlinear functions — Q7 [Hard] (ID: 7bbfbe70)

Question ID: 7bbfbe70

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A lab analyst observes a sample of a substance. An exponential model estimates that the mass, in grams, of the sample decreases by 24% every 22.96 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the lab analyst began observing the sample?

Answer

- A. $M = 100(0.24)^{t+22.96}$
- B. $M = 100(0.24)^{\frac{t}{22.96}}$
- C. $M = 100(0.76)^{t+22.96}$
- D. $M = 100(0.76)^{\frac{t}{22.96}}$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Identify the percentage decrease: the mass decreases by 24% every 22.96 minutes.
- Step 2: Convert this to a retention rate (the fraction of mass that REMAINS after each period): $100\% - 24\% = 76\%$, or as a decimal, 0.76.
- Step 3: Identify the time period for the exponent: since the decay happens 'every 22.96 minutes,' the exponent should be $t/22.96$, so that after exactly 22.96 minutes, the exponent equals 1 (one full decay period).
- Step 4: Assemble the model: $M = (\text{initial mass}) \times (\text{retention rate})^{(t / \text{period})} = 100(0.76)^{(t/22.96)}$.
- Step 5: This matches choice D.

Nonlinear functions — Q8 [Hard] (ID: a1e65979)

Question ID: a1e65979

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$ and passes through the points $(2, 33)$ and $(-1, 105)$. What is the value of $f(-2) - f(0)$?

Answer

- A. 81
- B. 129
- C. 192
- D. 225

Correct Answer: C

Step-by-Step Solution:

- Step 1: Write the quadratic in vertex form using the given vertex $(1, 9)$: $f(x) = a(x - 1)^2 + 9$.
- Step 2: Use the given point $(2, 33)$ to solve for a : substitute $x = 2$ and $f(x) = 33$: $33 = a(2 - 1)^2 + 9$, which simplifies to $33 = a(1) + 9$, so $33 = a + 9$.
- Step 3: Solve for a : $a = 33 - 9 = 24$.
- Step 4: Write the complete function: $f(x) = 24(x - 1)^2 + 9$.
- Step 5: (Optional check using the third point $(-1, 105)$): $f(-1) = 24(-1 - 1)^2 + 9 = 24(4) + 9 = 96 + 9 = 105$. This matches the given point, confirming the equation is correct.
- Step 6: Calculate $f(-2)$: $f(-2) = 24(-2 - 1)^2 + 9 = 24(-3)^2 + 9 = 24(9) + 9 = 216 + 9 = 225$.
- Step 7: Calculate $f(0)$: $f(0) = 24(0 - 1)^2 + 9 = 24(-1)^2 + 9 = 24(1) + 9 = 24 + 9 = 33$.
- Step 8: Calculate the requested difference: $f(-2) - f(0) = 225 - 33 = 192$.
- Step 9: This matches choice C.

Nonlinear functions — Q9 [Hard] (ID: 5acbd30)

Question ID: 5acbd30

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p ?

Answer

- A. 19
- B. 44
- C. 56
- D. 81

Correct Answer: D

Step-by-Step Solution:

Step 1: Identify the base of the exponential function: 0.19 (the number being raised to the power of x).

Step 2: Understand what multiplying by 0.19 means: each time x increases by 1, the new value is only 19% (0.19) of the previous value — meaning 19% of the value is RETAINED.

Step 3: Calculate the percent that is LOST (the percent decrease): $100\% - 19\% = 81\%$.

Step 4: Since $f(n)$ is $p\%$ less than $f(n-1)$, and we just calculated this percent decrease is 81%, we have $p = 81$.

Step 5: This matches choice D.

(Algebraic verification: $f(n)/f(n-1) = 56(0.19)^n / 56(0.19)^{n-1} = 0.19^1 = 0.19$. This ratio means $f(n) = 0.19 \times f(n-1)$, so $f(n)$ is 19% OF $f(n-1)$, meaning it decreased by $100\% - 19\% = 81\%$ from $f(n-1)$.)

Nonlinear functions — Q10 [Hard] [Grid-In] (ID: c95686e9)

Question ID: c95686e9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$h(t) = -16t^2 + b$$

The function h estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. The function estimates that the ball is 19.36 feet above the floor when it is released at $t = 0$. How many seconds after being released does the function estimate the ball will reach the floor?

Correct Answer: 1.1 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

Step 1: Use the given initial condition to find the constant b : at $t = 0$, the height is 19.36 feet, so substitute into $h(t) = -16t^2 + b$: $h(0) = -16(0)^2 + b = b$. Since $h(0) = 19.36$, this means $b = 19.36$.

Step 2: Write the complete function: $h(t) = -16t^2 + 19.36$.

Step 3: The ball reaches the floor when its height equals 0, so set $h(t) = 0$: $-16t^2 + 19.36 = 0$.

Step 4: Add $16t^2$ to both sides: $19.36 = 16t^2$.

Step 5: Divide both sides by 16: $t^2 = 19.36/16 = 1.21$.

Step 6: Take the square root of both sides: $t = \sqrt{1.21} = 1.1$ (using the positive root, since time elapsed must be a positive value).

Step 7: This is a grid-in question — enter 1.1 as your final answer.

Nonlinear functions — Q11 [Hard] (ID: 7bf77c19)

Question ID: 7bf77c19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The functions p and r are defined by the equations shown, where a and b are integer constants such that $a < b$ and $b < 0$. If $y = p(x)$ and $y = r(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

- I. $p(x) = a(4.1)^x + b$
II. $r(x) = a(4.1)^{x+b}$

Answer

- A. I only
B. II only
C. I and II
D. Neither I nor II

Correct Answer: D

Step-by-Step Solution:

- Step 1: For equation I, $p(x) = a(4.1)^x + b$, substitute $x = 0$: $p(0) = a(4.1)^0 + b$.
- Step 2: Simplify using the rule that anything to the power of 0 equals 1: $p(0) = a(1) + b = a + b$.
- Step 3: Notice that $a + b$ is a SUM of two separate constants — it's not simply 'a' or simply 'b' displayed alone; you'd need to actually calculate this sum to know the y -intercept's value. So equation I does NOT directly display its y -intercept as a single constant or coefficient.
- Step 4: For equation II, $r(x) = a(4.1)^{(x+b)}$, substitute $x = 0$: $r(0) = a(4.1)^{(0+b)} = a(4.1)^b$.
- Step 5: This is a PRODUCT of a and a power of 4.1 raised to b — again, not simply 'a' or simply 'b' by itself; you'd need to calculate this product to find the actual y -intercept value. So equation II also does NOT directly display its y -intercept.
- Step 6: Since neither equation shows its y -intercept as an already-visible single constant or coefficient, the correct answer is D: Neither I nor II.

Nonlinear equations in one variable and systems of equations in two variables

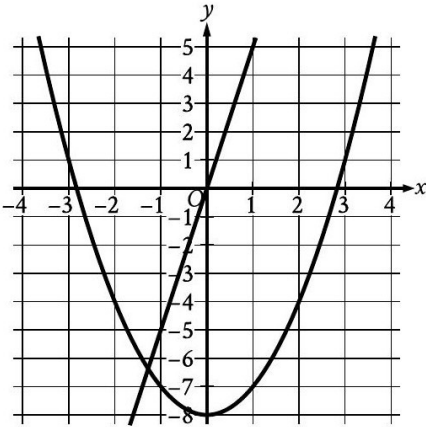
9 questions · Easy 1 / Medium 2 / Hard 6

Nonlinear equations in one variable and systems of equations in two variables — Q1 [Easy] (ID: f123e039)

Question ID: f123e039

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question



The graph of a system of a linear and a quadratic equation is shown. Which system of equations is represented by the graph?

Answer

- A. $y = -5x$
 $y = x^2 + 8$
- B. $y = -5x$
 $y = x^2 - 8$
- C. $y = 5x$
 $y = x^2 + 8$
- D. $y = 5x$
 $y = x^2 - 8$

Correct Answer: B

Step-by-Step Solution:

- Step 1: Identify the parabola's vertex directly from the graph: it appears to be at $(0, -8)$, the lowest point of the U-shaped curve.
- Step 2: Since a basic parabola $y = x^2$ shifted down by 8 units has equation $y = x^2 - 8$, and the vertex confirms this shift, the parabola's equation is $y = x^2 - 8$.
- Step 3: Identify a second clear point on the line: it appears to pass through the origin $(0, 0)$ and through approximately $(-1, 5)$.
- Step 4: Calculate the line's slope using these two points: $\text{slope} = (5 - 0) / (-1 - 0) = 5 / (-1) = -5$.
- Step 5: Since the line passes through the origin with slope -5 , its equation is $y = -5x$.
- Step 6: Compare both equations to the answer choices: $y = -5x$ and $y = x^2 - 8$ matches choice B exactly.

Question ID: 6bd40794

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$(x - 16)(x - 10)(x + 7)(x + 17) = 0$$

What is a positive solution to the given equation?

Correct Answer: 16 (or 10) (Student-Produced Response — no answer choices given)

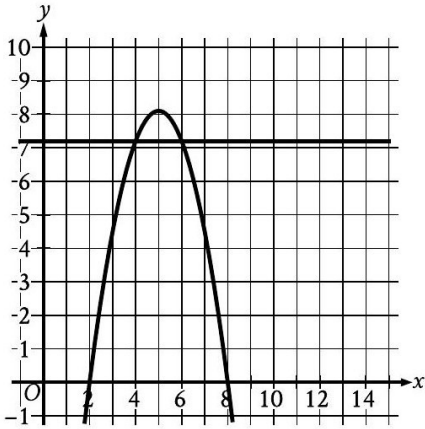
Step-by-Step Solution:

- Step 1: Recognize that the equation is already fully factored: $(x - 16)(x - 10)(x + 7)(x + 17) = 0$.
- Step 2: Apply the Zero Product Property: since the product of these four factors equals 0, at least one of the factors itself must equal 0.
- Step 3: Set each factor equal to 0 individually and solve: $x - 16 = 0$ gives $x = 16$; $x - 10 = 0$ gives $x = 10$; $x + 7 = 0$ gives $x = -7$; $x + 17 = 0$ gives $x = -17$.
- Step 4: The complete solution set is $\{16, 10, -7, -17\}$. The question asks for a POSITIVE solution — both 16 and 10 qualify.
- Step 5: This is a grid-in question — enter either 16 or 10 as your final answer; both are accepted as correct.

Question ID: 5f10c095

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question



The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y) . What is a possible value of x ?

Answer

- A. 4
- B. 5
- C. 7.2
- D. 8

Correct Answer: A

Step-by-Step Solution:

- Step 1: Identify the horizontal line's equation from the graph: it appears to be a flat line at $y = 7$.
- Step 2: Look for where this horizontal line crosses the parabola — these crossing points are the solutions to the system.
- Step 3: Reading the graph carefully, the line and parabola appear to intersect at approximately $x = 4$ and $x = 6$ (the parabola's vertex is at approximately $(5, 8)$, and the two intersection points are symmetric around this vertex).
- Step 4: Compare these x -values to the answer choices: $x = 4$ is listed as choice A, confirming it as a valid solution.

Question ID: 46308566

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$r^2 + qr = 2r - 55$$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q ?

Correct Answer: 16 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Start with the given equation: $r^2 + qr = 2r - 55$.
- Step 2: Move all terms to one side to get standard form: $r^2 + qr - 2r + 55 = 0$.
- Step 3: Combine the like terms involving r : $r^2 + (q - 2)r + 55 = 0$.
- Step 4: Identify the coefficients for the discriminant formula: $a = 1$, $b = (q - 2)$, $c = 55$.
- Step 5: For no real solutions, set up the discriminant inequality: $b^2 - 4ac < 0$, which becomes $(q - 2)^2 - 4(1)(55) < 0$.
- Step 6: Simplify: $(q - 2)^2 - 220 < 0$, so $(q - 2)^2 < 220$.
- Step 7: Take the square root of both sides (remembering this creates a two-sided inequality): $-\sqrt{220} < q - 2 < \sqrt{220}$.
- Step 8: Calculate $\sqrt{220} \approx 14.83$.
- Step 9: Add 2 to all three parts of the inequality: $2 - 14.83 < q < 2 + 14.83$, which gives approximately $-12.83 < q < 16.83$.
- Step 10: Since q is stated to be an INTEGER constant, find the largest integer within this range: the largest integer less than 16.83 is 16.
- Step 11: This is a grid-in question — enter 16 as your final answer.

Question ID: e833de9a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$x^2 + y^2 = 36$$

$$y = mx + \frac{b}{4}$$

In the given system of equations, m and b are negative constants. In the xy -plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

Answer

A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Substitute the known x-coordinate ($x = -5$) into the CIRCLE equation to find y: $(-5)^2 + y^2 = 36$.
- Step 2: Simplify: $25 + y^2 = 36$.
- Step 3: Solve for y^2 : $y^2 = 36 - 25 = 11$.
- Step 4: Take the square root of both sides: $y = \pm\sqrt{11}$. Since the problem states $y < 0$, choose the negative root: $y = -\sqrt{11}$.
- Step 5: Now substitute both $x = -5$ and $y = -\sqrt{11}$ into the LINE equation: $y = mx + b/4$, giving $-\sqrt{11} = m(-5) + b/4$.
- Step 6: Simplify the right side: $-\sqrt{11} = -5m + b/4$.
- Step 7: Add $5m$ to both sides to isolate the $b/4$ term: $-\sqrt{11} + 5m = b/4$, or equivalently $5m - \sqrt{11} = b/4$.
- Step 8: Multiply both sides by 4 to solve for b: $b = 4(5m - \sqrt{11}) = 20m - 4\sqrt{11}$.
- Step 9: This matches choice D.

Question ID: 5da5c665

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$4x^2 - 5x - 5 = 0$$

What is the greatest solution to the given equation?

Answer

- A. $\frac{5}{8} - \frac{\sqrt{105}}{8}$
- B. $\frac{5}{8} + \frac{\sqrt{105}}{8}$
- C. $\frac{5}{4} - \frac{\sqrt{105}}{4}$
- D. $\frac{5}{4} + \frac{\sqrt{105}}{4}$

Correct Answer: B

Step-by-Step Solution:

- Step 1: Identify the coefficients from the standard form $ax^2 + bx + c = 0$: $a = 4$, $b = -5$, $c = -5$.
- Step 2: Write out the quadratic formula: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a)$.
- Step 3: Substitute the coefficients: $x = [-(-5) \pm \sqrt{((-5)^2 - 4(4)(-5))}] / (2(4))$.
- Step 4: Simplify the numerator's first term: $-(-5) = 5$.
- Step 5: Simplify inside the square root: $(-5)^2 = 25$, and $4(4)(-5) = -80$, so subtracting a negative becomes addition: $25 - (-80) = 25 + 80 = 105$.
- Step 6: Simplify the denominator: $2(4) = 8$.
- Step 7: The formula now reads: $x = [5 \pm \sqrt{105}] / 8$.
- Step 8: Since the question asks for the GREATEST solution, use the '+' version: $x = 5/8 + \sqrt{105}/8$.
- Step 9: This matches choice B.

Question ID: c8db0e19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{1}{4xy} + xyz = \frac{1}{3yz}$$

In the given equation, x , y , and z are positive numbers. Which expression is equivalent to y ?

Answer

- A. $\frac{4x-3z}{12x^2z^2}$
- B. $\sqrt{\frac{4x-3z}{12x^2z^2}}$
- C. $\frac{1}{3xz^2-4x^2z}$
- D. $\sqrt{\frac{1}{3xz^2-4x^2z}}$

Correct Answer: B

Step-by-Step Solution:

- Step 1: Start with the given equation: $1/(4xy) + xyz = 1/(3yz)$.
- Step 2: Multiply every single term in the equation by y to begin clearing y from the denominators: $y/(4xy) + xy(yz) = y/(3yz)$.
- Step 3: Simplify each term: $y/(4xy) = 1/(4x)$; $xy(yz) = xy^2z$; $y/(3yz) = 1/(3z)$. The equation becomes: $1/(4x) + xy^2z = 1/(3z)$.
- Step 4: Subtract $1/(4x)$ from both sides to isolate the xy^2z term: $xy^2z = 1/(3z) - 1/(4x)$.
- Step 5: Combine the right side into a single fraction using the common denominator $12xz$: $1/(3z) = 4x/(12xz)$, and $1/(4x) = 3z/(12xz)$, so $1/(3z) - 1/(4x) = (4x - 3z)/(12xz)$.
- Step 6: Now the equation reads: $xy^2z = (4x - 3z)/(12xz)$.
- Step 7: Divide both sides by xz to isolate y^2 : $y^2 = (4x - 3z)/(12xz \times xz) = (4x - 3z)/(12x^2z^2)$.
- Step 8: Take the square root of both sides. Since y is stated to be a positive number, keep only the positive root: $y = \sqrt{[(4x - 3z)/(12x^2z^2)]}$.
- Step 9: This matches choice B.

Nonlinear equations in one variable and systems of equations in two variables — Q8 [Hard] (ID: 9a1f1941)

Question ID: 9a1f1941

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$|4x - 3| = -9$

How many solutions does the given equation have?

Answer

- A. Zero
- B. One
- C. Two
- D. Infinitely many

Correct Answer: A

Step-by-Step Solution:

- Step 1: Look at the given equation: $|4x - 3| = -9$.
- Step 2: Recall the fundamental property of absolute value: $|anything|$ is always greater than or equal to 0 — it can never be negative, because absolute value measures distance from zero, and distance is never negative.
- Step 3: Since the right side of the equation is -9 (a negative number), and the left side (an absolute value expression) can never be negative, there is no possible value of x that could make this equation true.
- Step 4: Conclude that the equation has zero solutions, matching choice A.

Question ID: 30596346

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{1}{7}(x - k)(16 - x^2) = 0$$

In the given equation, k is a positive constant. The sum of the solutions to the equation is 59. What is the value of k ?

Correct Answer: 59 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Recognize the equation is already factored: $(1/7)(x - k)(16 - x^2) = 0$.
- Step 2: Since $1/7$ is just a nonzero constant multiplying everything, it doesn't affect where the equation equals zero — focus on the two remaining factors: $(x - k)$ and $(16 - x^2)$.
- Step 3: Set the first factor equal to zero: $x - k = 0$, giving $x = k$.
- Step 4: Set the second factor equal to zero: $16 - x^2 = 0$, which means $x^2 = 16$.
- Step 5: Take the square root of both sides: $x = 4$ or $x = -4$ (remembering that squaring EITHER a positive or negative 4 gives 16, so BOTH are valid solutions).
- Step 6: The complete solution set is $\{k, 4, -4\}$.
- Step 7: Add up all the solutions and set this equal to the given sum: $k + 4 + (-4) = 59$.
- Step 8: Simplify: $k + 0 = 59$, so $k = 59$.
- Step 9: This is a grid-in question — enter 59 as your final answer.

Equivalent expressions

7 questions · Easy 1 / Medium 2 / Hard 4

Equivalent expressions — Q1 [Easy] (ID: 91aa9aa9)

Question ID: 91aa9aa9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $(9x^2 + 6) - (8x^2)$?

Answer

- A. 7
- B. $17x^2 + 6$
- C. $x^2 - 6$
- D. $x^2 + 6$

Correct Answer: D

Step-by-Step Solution:
Step 1: Write out the given expression: $(9x^2 + 6) - (8x^2)$.
Step 2: Distribute the subtraction sign across the second parentheses: $9x^2 + 6 - 8x^2$.
Step 3: Identify the like terms: $9x^2$ and $-8x^2$ are like terms (both have x^2), while 6 is a constant with no matching term to combine.
Step 4: Combine the like terms: $9x^2 - 8x^2 = x^2$.
Step 5: Write the final simplified expression: $x^2 + 6$.
Step 6: This matches choice D.

Equivalent expressions — Q2 [Medium] (ID: 04887c7b)

Question ID: 04887c7b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$$f(x) = 2x + 3$$

$$g(x) = 7x - 2$$

$$h(x) = 5x + 6$$

The functions f , g , and h are defined as shown. If $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a , b , and c are constants, what is the value of b ?

Answer

- A. -5
- B. 12
- C. 20
- D. 22

Correct Answer: B

Step-by-Step Solution:

- Step 1: Write out $f(x) \cdot g(x)$ as a multiplication: $(2x + 3)(7x - 2)$.
- Step 2: Use FOIL (First, Outer, Inner, Last) to expand this product: First: $2x \times 7x = 14x^2$. Outer: $2x \times (-2) = -4x$. Inner: $3 \times 7x = 21x$. Last: $3 \times (-2) = -6$.
- Step 3: Combine these four terms: $14x^2 - 4x + 21x - 6$.
- Step 4: Combine the like x -terms: $-4x + 21x = 17x$. So $f(x) \cdot g(x) = 14x^2 + 17x - 6$.
- Step 5: Now subtract $h(x) = 5x + 6$ from this result: $(14x^2 + 17x - 6) - (5x + 6)$.
- Step 6: Distribute the negative sign across $h(x)$: $14x^2 + 17x - 6 - 5x - 6$.
- Step 7: Combine like terms: the x -terms: $17x - 5x = 12x$. The constants: $-6 - 6 = -12$.
- Step 8: The fully simplified expression is: $14x^2 + 12x - 12$.
- Step 9: Comparing this to the general form $ax^2 + bx + c$, we can see $b = 12$ (the coefficient of the x term).
- Step 10: This matches choice B.

Equivalent expressions — Q3 [Medium] (ID: f8871ccd)

Question ID: f8871ccd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which expression is equivalent to $2\left(x - \frac{9}{2}\right)(x + 8)$?

Answer

- A. $2x^2 - 72$
- B. $2x^2 + 7x - 72$
- C. $2x^2 - x - 36$
- D. $x^2 + 7x - 36$

Correct Answer: B

Step-by-Step Solution:

- Step 1: Start with the given expression: $2(x - 9/2)(x + 8)$.
- Step 2: Distribute the leading 2 into the first binomial to clear the fraction: $2 \times x = 2x$, and $2 \times (-9/2) = -9$. This gives $(2x - 9)$ for the first factor.
- Step 3: Rewrite the expression with this simplified first factor: $(2x - 9)(x + 8)$.
- Step 4: Use FOIL to expand: First: $2x \times x = 2x^2$. Outer: $2x \times 8 = 16x$. Inner: $-9 \times x = -9x$. Last: $-9 \times 8 = -72$.
- Step 5: Combine these four terms: $2x^2 + 16x - 9x - 72$.
- Step 6: Combine the like x-terms: $16x - 9x = 7x$.
- Step 7: The fully expanded expression is: $2x^2 + 7x - 72$.
- Step 8: This matches choice B.

Equivalent expressions — Q4 [Hard] [Grid-In] (ID: 38c90632)

Question ID: 38c90632

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $\frac{29x+102}{x(x+51)}$ is equivalent to $\frac{p}{x} + \frac{w}{x+51}$, where p and w are constants. What is the value of w ?

Correct Answer: 27 (Student-Produced Response — no answer choices given)**Step-by-Step Solution:**

Step 1: Write the right side of the given equivalence, $p/x + w/(x+51)$, as a single combined fraction using the common denominator $x(x+51)$: $[p(x+51) + w(x)] / [x(x+51)]$.

Step 2: Expand the numerator: $p(x+51) = px + 51p$, so the full numerator is $px + 51p + wx$.

Step 3: Combine the like terms in the numerator (the x -terms): $px + wx = (p+w)x$. The numerator becomes $(p+w)x + 51p$.

Step 4: Since this combined fraction must equal the original expression $(29x+102)/[x(x+51)]$, and the denominators already match, set the numerators equal to each other: $(p+w)x + 51p = 29x + 102$.

Step 5: Match the coefficients of x on both sides: $p + w = 29$.

Step 6: Match the constant terms on both sides: $51p = 102$.

Step 7: Solve the constant-term equation for p : $p = 102/51 = 2$.

Step 8: Substitute $p = 2$ into the x -coefficient equation: $2 + w = 29$.

Step 9: Solve for w : $w = 29 - 2 = 27$.

Step 10: This is a grid-in question — enter 27 as your final answer.

Equivalent expressions — Q5 [Hard] [Grid-In] (ID: 00165291)

Question ID: 00165291

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

Correct Answer: 31/2 or 15.5 (Student-Produced Response — no answer choices given)**Step-by-Step Solution:**

Step 1: Notice that $6x^4 + 31x^2 + 35$ is a 'quadratic in disguise' — if you let $y = x^2$, the expression becomes $6y^2 + 31y + 35$, a standard quadratic.

Step 2: Find the roots of $6y^2 + 31y + 35 = 0$ using the quadratic formula: $y = [-31 \pm \sqrt{31^2 - 4(6)(35)}] / (2 \times 6) = [-31 \pm \sqrt{961 - 840}] / 12 = [-31 \pm \sqrt{121}] / 12 = [-31 \pm 11] / 12$.

Step 3: Calculate both roots: $y = (-31+11)/12 = -20/12 = -5/3$, and $y = (-31-11)/12 = -42/12 = -7/2$.

Step 4: This means $6y^2 + 31y + 35 = 6(y + 5/3)(y + 7/2)$, which can be rewritten (redistributing the 6) as $(3y + 5)(2y + 7)$ — substituting back $y = x^2$ gives the INTEGER factorization $(3x^2 + 5)(2x^2 + 7)$, so $a = 5$ and $b = 7$.

Step 5: For the SECOND (non-integer) factorization, redistribute the SAME two roots differently between the two binomials: instead of matching $5/3$ with the '3' coefficient, match $7/2$ with the '3' coefficient instead: $(3x^2 + 21/2)(2x^2 + 10/3)$ — here, $c/3 = 7/2$ gives $c = 21/2$, and $d/2 = 5/3$ gives $d = 10/3$, both non-integers as required.

Step 6: Now calculate the requested value: $a + c = 5 + 21/2$.

Step 7: Convert 5 to a fraction with denominator 2: $5 = 10/2$. So $a + c = 10/2 + 21/2 = 31/2$.

Step 8: This is a grid-in question — enter 31/2 or its decimal equivalent, 15.5, as your final answer.

Equivalent expressions — Q6 [Hard] (ID: efc469c4)

Question ID: efc469c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$-14(5x - 2)^2 + 6(5x - 3)^2$$

The given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, where a , b , and c are constants. What is the value of $a + b + c$?

Answer

- A. -612
- B. -306
- C. -102
- D. -17

Correct Answer: A

Step-by-Step Solution:

- Step 1: Expand the first squared binomial: $(5x-2)^2 = (5x)^2 - 2(5x)(2) + 2^2 = 25x^2 - 20x + 4$.
- Step 2: Expand the second squared binomial: $(5x-3)^2 = (5x)^2 - 2(5x)(3) + 3^2 = 25x^2 - 30x + 9$.
- Step 3: Distribute the leading coefficient -14 across the first expansion: $-14(25x^2 - 20x + 4) = -350x^2 + 280x - 56$.
- Step 4: Distribute the leading coefficient 6 across the second expansion: $6(25x^2 - 30x + 9) = 150x^2 - 180x + 54$.
- Step 5: Add these two results together, combining like terms: x^2 terms: $-350x^2 + 150x^2 = -200x^2$. x terms: $280x - 180x = 100x$. Constants: $-56 + 54 = -2$.
- Step 6: The combined expression is: $-200x^2 + 100x - 2$.
- Step 7: Set this equal to the target form $(a/6)x^2 + (b/6)x + c/6$, meaning $a/6 = -200$, $b/6 = 100$, and $c/6 = -2$.
- Step 8: Solve for each constant by multiplying both sides by 6: $a = -200 \times 6 = -1200$; $b = 100 \times 6 = 600$; $c = -2 \times 6 = -12$.
- Step 9: Calculate the requested sum: $a + b + c = -1200 + 600 + (-12) = -612$.
- Step 10: This matches choice A.

Equivalent expressions — Q7 [Hard] (ID: 2289b199)

Question ID: 2289b199

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\frac{\sqrt[3]{n^{17}p^5}}{5n^2(\sqrt[8]{p^8})}$$

For all positive values of n and p , the given expression is equivalent to which of the following?

I. $\frac{(n^{\frac{14}{3}})(p^{\frac{5}{3}})}{5np}$

II. $\frac{\sqrt[3]{n^{11}p^2}}{5}$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: C

Step-by-Step Solution:

- Step 1: Rewrite the numerator using fractional exponent notation: $\sqrt[3]{(n^{17}p^5)} = (n^{17}p^5)^{\frac{1}{3}} = n^{\frac{17}{3}}p^{\frac{5}{3}}$.
- Step 2: Rewrite the denominator's radical term: $\sqrt[8]{(p^8)} = p^{\frac{8}{8}} = p^1 = p$ (since 8 divided by 8 equals exactly 1).
- Step 3: Write the full original expression using these simplified forms: $[n^{\frac{17}{3}}p^{\frac{5}{3}}] / [5n^2p]$.
- Step 4: Simplify by subtracting exponents for matching bases (dividing powers with the same base means subtracting exponents): for n : $\frac{17}{3} - 2 = \frac{17}{3} - \frac{6}{3} = \frac{11}{3}$. For p : $\frac{5}{3} - 1 = \frac{5}{3} - \frac{3}{3} = \frac{2}{3}$.
- Step 5: The fully simplified original expression is: $(\frac{1}{5}) n^{\frac{11}{3}} p^{\frac{2}{3}}$.
- Step 6: Check option I: $[n^{\frac{14}{3}}p^{\frac{5}{3}}] / (5np)$. Simplify exponents: for n : $\frac{14}{3} - 1 = \frac{14}{3} - \frac{3}{3} = \frac{11}{3}$. For p : $\frac{5}{3} - 1 = \frac{5}{3} - \frac{3}{3} = \frac{2}{3}$. This gives $(\frac{1}{5})n^{\frac{11}{3}}p^{\frac{2}{3}}$ — an EXACT match to Step 5.
- Step 7: Check option II: $\sqrt[3]{(n^{11}p^2)}/5 = (n^{11}p^2)^{\frac{1}{3}}/5 = n^{\frac{11}{3}}p^{\frac{2}{3}}/5$ — also an EXACT match to Step 5.
- Step 8: Since BOTH option I and option II are algebraically equivalent to the original expression, the correct answer is C: I and II.

Problem-Solving and Data Analysis

Math · 19 questions · Εκφώνηση / Correct Answer / Step-by-Step Solution

Ratios, rates, proportional relationships, and units

3 questions · Easy 1 / Medium 0 / Hard 2

Ratios, rates, proportional relationships, and units — Q1 [Easy] (ID: de799680)

Question ID: de799680

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Easy

Question

A cluster fly is traveling at a velocity of 76 centimeters per second. What is this velocity, in **millimeters** per second?
(1 centimeter = 10 millimeters)

Answer

- A. 7,600
- B. 836
- C. 760
- D. 86

Correct Answer: C

Step-by-Step Solution:

- Step 1: Identify the given rate: 76 centimeters per second.
- Step 2: Identify the conversion factor: 1 centimeter = 10 millimeters.
- Step 3: Since millimeters are a SMALLER unit than centimeters, converting to millimeters should give a LARGER number — this means multiplying, not dividing.
- Step 4: Multiply: $76 \text{ cm/s} \times 10 \text{ mm/cm} = 760 \text{ mm/s}$ (the 'cm' units cancel out, leaving 'mm').
- Step 5: This matches choice C.

Ratios, rates, proportional relationships, and units — Q2 [Hard] (ID: 4c5ea142)

Question ID: 4c5ea142

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

On average, a certain plant grows 87 millimeters every m months. At this rate, which expression represents the number of millimeters, on average, the plant grows every k years?

Answer

- A. $\frac{29m}{4k}$
- B. $\frac{29k}{4m}$
- C. $\frac{1,044m}{k}$
- D. $\frac{1,044k}{m}$

Correct Answer: D

Step-by-Step Solution:

Step 1: Write the given growth rate as a fraction: 87 millimeters per m months, or $87/m$ mm/month.

Step 2: Convert this monthly rate into a YEARLY rate by multiplying by 12 (since there are 12 months in one year): $(87/m) \times 12 = 1044/m$ mm per year.

Step 3: To find the total growth over k years (not just one year), multiply this yearly rate by k : $(1044/m) \times k = 1044k/m$.

Step 4: This matches choice D.

Ratios, rates, proportional relationships, and units — Q3 [Hard] (ID: 1b96c116)

Question ID: 1b96c116

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

A rectangle has an area of 3,168 square inches. What is the area, in **square feet**, of this rectangle? (1 foot = 12 inches)

Answer

- A. 22
- B. 56.3
- C. 264
- D. 675.4

Correct Answer: A

Step-by-Step Solution:

Step 1: Recall the linear conversion: 1 foot = 12 inches.

Step 2: Since area involves TWO dimensions (length $\times$ width), square the conversion factor to get the area conversion: 1 square foot = $12^2 = 144$ square inches.

Step 3: To convert the given area from square inches to square feet, divide by 144 (since square inches is the smaller unit, converting to the larger unit square feet requires dividing): $3,168 \div 144 = 22$.

Step 4: This matches choice A.

Percentages

4 questions · Easy 0 / Medium 0 / Hard 4

Percentages — Q1 [Hard] (ID: 28ff3966)

Question ID: 28ff3966

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

A square scale drawing has a side length of 55 inches, and 1 inch on the drawing represents an actual distance of 19 miles. A larger version of the same drawing is printed as a square with the side length 65% longer than the side length of the previous drawing. On the larger drawing, which of the following is closest to the actual distance, in miles, represented by 1 inch?

Answer

- A. 6.65
- B. 11.52
- C. 31.35
- D. 35.75

Correct Answer: B

Step-by-Step Solution:

- Step 1: Calculate the total actual distance represented by the original drawing's side: $55 \text{ inches} \times 19 \text{ miles/inch} = 1,045 \text{ miles}$.
- Step 2: Recognize that this actual distance (1,045 miles) stays the SAME even when the drawing is resized — only the drawing's size changes, not the real place it represents.
- Step 3: Calculate the new drawing's side length: the new side is 65% LONGER than the original 55 inches, so multiply by 1.65 ($100\% + 65\% = 165\%$, or 1.65 as a decimal): $55 \times 1.65 = 90.75 \text{ inches}$.
- Step 4: Since the new 90.75-inch side still represents the same 1,045 actual miles, calculate the new scale by dividing: $1,045 \div 90.75 \approx 11.5152 \text{ miles per inch}$.
- Step 5: Rounding to match the answer choices, this is closest to 11.52, matching choice B.

Percentages — Q2 [Hard] (ID: 8bb94668)

Question ID: 8bb94668

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is 5,900% of the speed of object C and the speed of object C is 0.008% of the speed of object B, which expression represents the value of $a + b$ in terms of c ?

Answer

- A. $1,840c$
- B. $5,908c$
- C. $6,025c$
- D. $12,559c$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Convert 'a is 5,900% of c' into an equation: 5,900% as a decimal is $5900/100 = 59$, so $a = 59c$.
- Step 2: Convert 'c is 0.008% of b' into an equation: 0.008% as a decimal is $0.008/100 = 0.00008$, so $c = 0.00008b$.
- Step 3: Solve this second equation for b (since we need b in terms of c to add it to a): $b = c / 0.00008$.
- Step 4: Calculate this division: $1/0.00008 = 12,500$, so $b = 12,500c$.
- Step 5: Now add a and b, both now expressed in terms of c: $a + b = 59c + 12,500c$.
- Step 6: Combine like terms: $59c + 12,500c = 12,559c$.
- Step 7: This matches choice D.

Percentages — Q3 [Hard] (ID: 6feae9c3)

Question ID: 6feae9c3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

At a recreation center, visitors used the exercise room for a total of t hours in July. At the same recreation center, visitors used the exercise room for 3.49t hours in August. What is the percent increase in the number of hours visitors used the exercise room from July to August?

Answer

- A. 2.49%
- B. 3.49%
- C. 249%
- D. 349%

Correct Answer: C

Step-by-Step Solution:

- Step 1: Identify the original value (July hours): t .
- Step 2: Identify the new value (August hours): $3.49t$.
- Step 3: Calculate the amount of increase: new value – original value = $3.49t - t = 2.49t$.
- Step 4: Apply the percent increase formula: (increase / original value) $\times 100\%$.
- Step 5: Substitute the values: $(2.49t / t) \times 100\%$.
- Step 6: The t in the numerator and denominator cancels out completely: $2.49 \times 100\% = 249\%$.
- Step 7: This matches choice C.

Percentages — Q4 [Hard] [Grid-In] (ID: 98818acf)

Question ID: 98818acf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard ■■■■■■■■■■

Question

The number of audiobooks in a library increased by 131% from last year to this year. The number of audiobooks in the library last year was n , and the number of audiobooks in the library this year is bn , where b and n are constants. What is the value of b ?

Correct Answer: 2.31 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

Step 1: Understand what 'increased by 131%' means: the new amount equals the original amount (100%) PLUS an additional 131% increase, for a total of $100\% + 131\% = 231\%$.

Step 2: Convert 231% into a decimal multiplier: $231/100 = 2.31$.

Step 3: Since the new amount is bn (b times the original n), and we just determined the new amount is 2.31 times the original, this means $b = 2.31$.

Step 4: This is a grid-in question — enter 2.31 as your final answer.

One-variable data: distributions and measures of center and spread

4 questions · Easy 1 / Medium 1 / Hard 2

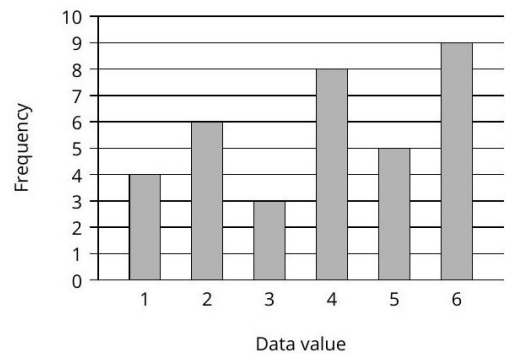
One-variable data: distributions and measures of center and spread — Q1 [Easy] [Grid-In] (ID: 559476f4)

Question ID: 559476f4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

The bar graph shows the frequency of each data value in a data set.



What is the frequency of data value 2 in this data set?

Correct Answer: 6 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Locate '2' on the horizontal (data value) axis.
- Step 2: Trace straight up from that point until you reach the top of the bar directly above it.
- Step 3: From the top of that bar, trace straight across (left) to the vertical (frequency) axis to read its height.
- Step 4: The bar for data value 2 reaches a height of 6.
- Step 5: This is a grid-in question — enter 6 as your final answer.

One-variable data: distributions and measures of center and spread — Q2 [Medium] (ID: d73a67d7)

Question ID: d73a67d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Medium

Question

Which of the following lists represents a data set with the smallest standard deviation?

Answer

- A. 51, 52, 54, 56, 57
- B. 52, 52, 54, 56, 56
- C. 52, 53, 54, 55, 56
- D. 53, 54, 54, 54, 55

Correct Answer: D

Step-by-Step Solution:

- Step 1: Calculate (or estimate) the mean of each list — notice all four lists happen to have a mean of 54.
- Step 2: Since all means are equal, compare how far each list's individual values deviate from 54 — the list with the SMALLEST deviations overall will have the smallest standard deviation.
- Step 3: List A (51,52,54,56,57): deviations of $-3,-2,0,2,3$ — fairly spread out.
- Step 4: List B (52,52,54,56,56): deviations of $-2,-2,0,2,2$ — moderately spread out.
- Step 5: List C (52,53,54,55,56): deviations of $-2,-1,0,1,2$ — more tightly clustered.
- Step 6: List D (53,54,54,54,55): deviations of $-1,0,0,0,1$ — extremely tightly clustered, with three values exactly AT the mean.
- Step 7: List D has by far the smallest deviations from the mean, meaning it has the smallest standard deviation.
- Step 8: This matches choice D.

One-variable data: distributions and measures of center and spread — Q3 [Hard] (ID: f6c18a66)

Question ID: f6c18a66

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of 11 days.

Deliveries	Days
0	2
3	2
4	2
5	2
6	1
7	1
13	1

The data value 13 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 10 data values. Which statement best compares the median of data set A and the median of data set B?

Answer

- A. The median of data set B is less than the median of data set A.
- B. The median of data set B is greater than the median of data set A.
- C. The median of data set B is equal to the median of data set A.
- D. There is not enough information to compare the medians of the two data sets.

Correct Answer: C

Step-by-Step Solution:

Step 1: 'Unroll' the frequency table into a full ordered list for data set A: two 0's, two 3's, two 4's, two 5's, one 6, one 7, one 13 — giving: 0, 0, 3, 3, 4, 4, 5, 5, 6, 7, 13 (11 values total).

Step 2: Find the median of data set A: with 11 values (an odd number), the median is the middle (6th) value. Counting through the ordered list: 1st=0, 2nd=0, 3rd=3, 4th=3, 5th=4, 6th=4. The median of set A is 4.

Step 3: Create data set B by removing the erroneous value 13: 0, 0, 3, 3, 4, 4, 5, 5, 6, 7 (10 values total).

Step 4: Find the median of data set B: with 10 values (an even number), the median is the average of the two middle values (the 5th and 6th). Counting through: 5th=4, 6th=4. The median of set B is $(4+4)/2 = 4$.

Step 5: Compare the two medians: both data set A and data set B have a median of 4 — they are EQUAL.

Step 6: This matches choice C.

One-variable data: distributions and measures of center and spread — Q4 [Hard] (ID: cf2be18d)

Question ID: cf2be18d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

$a, 26, 29, b, 31, 47, c$

For the given data set, the data values are listed in ascending order, where a , b , and c are constants. For this data set, the mean is 36, the median is 29, and the range is 72. What is the value of c ?

Answer

- A. 54
- B. 72
- C. 81
- D. 98

Correct Answer: C

Step-by-Step Solution:

Step 1: Since the 7 values are listed in ascending order ($a, 26, 29, b, 31, 47, c$), identify the median's position: for 7 values, the median is the 4th (middle) value, which is b .

Step 2: Since the median is given as 29, set $b = 29$.

Step 3: Use the mean to find the sum of all values: $\text{mean} \times \text{count} = \text{sum}$, so $36 \times 7 = 252$.

Step 4: Substitute all known values into the sum equation: $a + 26 + 29 + 29 + 31 + 47 + c = 252$.

Step 5: Add up the known numbers: $26 + 29 + 29 + 31 + 47 = 162$. The equation becomes: $a + c + 162 = 252$.

Step 6: Solve for $a + c$: $a + c = 252 - 162 = 90$.

Step 7: Use the range (largest minus smallest): since the list is ascending, the range is $c - a = 72$.

Step 8: Now solve this system of two equations: $a + c = 90$ and $c - a = 72$. Add both equations together: $(a+c) + (c-a) = 90 + 72$, which simplifies to $2c = 162$.

Step 9: Divide both sides by 2: $c = 81$.

Step 10: This matches choice C.

Two-variable data: models and scatterplots

3 questions · Easy 1 / Medium 2 / Hard 0

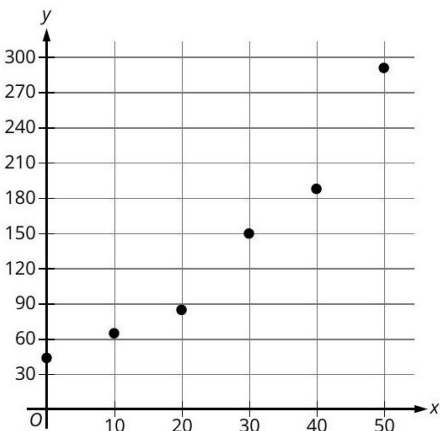
Two-variable data: models and scatterplots — Q1 [Easy] (ID: dbd89c59)

Question ID: dbd89c59

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Easy

Question

The scatterplot shows the relationship between the number of swans, y , in a certain population and the number of years since 1970, x , where $0 \leq x \leq 50$.



Which of the following equations is the most appropriate exponential model for the data shown in the scatterplot?

Answer

A. $y = 43(1.04)^{-x}$

B. $y = 43(1.04)^x$

C. $y = 288(1.04)^{-x}$

D. $y = 288(1.04)^x$

Correct Answer: B

Step-by-Step Solution:

Step 1: Read the y -intercept directly from the scatterplot: the leftmost point (at $x = 0$) sits at approximately $y = 43$.

Step 2: Observe the overall trend of the data: as x increases from 0 to 50, y clearly increases (grows) from about 43 up to nearly 290 — this is GROWTH, not decay.

Step 3: Match these observations to the exponential model form $y = a(b)^x$: the starting value a should be approximately 43, and since the data is growing, the exponent should be POSITIVE x (not negative x , which would represent decay).

Step 4: This matches choice B: $y = 43(1.04)^x$.

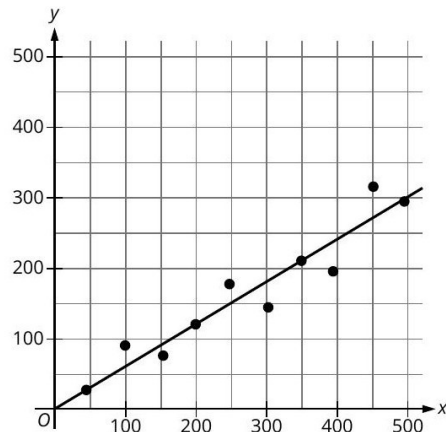
Two-variable data: models and scatterplots — Q2 [Medium] (ID: b3547a10)

Question ID: b3547a10

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

A group of 10 botanists recorded data on the germination rates of a certain type of seed for one growing season. The scatterplot shows the relationship between the number of seeds planted, x , and the number of seeds that germinated, y , for each of the botanists. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit in this context?

Answer

- A. The number of seeds planted is predicted to increase by 60 seeds every 100 days.
- B. The number of seeds planted is predicted to increase by 300 seeds every 100 days.
- C. The number of seeds that germinate is predicted to increase by 60 seeds for every additional 100 seeds that are planted.
- D. The number of seeds that germinate is predicted to increase by 300 seeds for every additional 100 seeds that are planted.

Correct Answer: C

Step-by-Step Solution:

- Step 1: Identify which variable is on the x -axis (independent/input variable) and which is on the y -axis (dependent/output variable): x = number of seeds planted, y = number of seeds that germinated.
- Step 2: Estimate two points on the line of best fit from the graph: approximately $(0, 30)$ and $(500, 310)$.
- Step 3: Calculate the slope using these two points: $\text{slope} = (310 - 30) / (500 - 0) = 280/500 = 0.56$.
- Step 4: Compare this slope to the answer choices, which are given as 'X seeds per 100 seeds': 0.56 is closest to $60/100 = 0.60$.
- Step 5: Since the slope relates y (germinated) to x (planted), the correct interpretation must describe germinated seeds increasing per planted seeds — not the reverse.
- Step 6: This matches choice C: 'The number of seeds that germinate is predicted to increase by 60 seeds for every additional 100 seeds that are planted.'

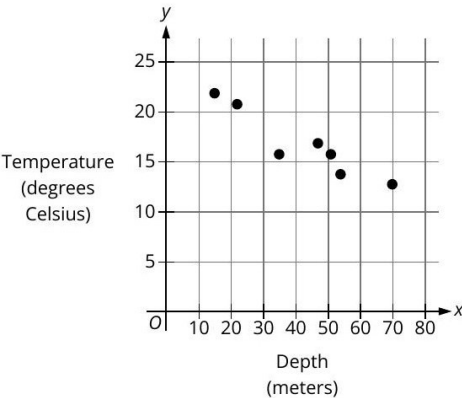
Two-variable data: models and scatterplots — Q3 [Medium] (ID: 70a2776f)

Question ID: 70a2776f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

As part of a study of lake conditions, the water temperature was recorded at different depths below the surface of a certain lake. The data shown in the scatterplot give the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of the lake.



Which of the following is closest to the slope of a line of best fit for the data shown?

Answer

- A. -0.17
- B. -2.17
- C. -6.02
- D. -8.02

Correct Answer: A

Step-by-Step Solution:

- Step 1: Choose two points that represent the overall trend of the scattered data — one from early depths and one from later depths. Reading the graph, approximate points are (15, 22) and (70, 13).
- Step 2: Calculate the change in y (temperature): $13 - 22 = -9$.
- Step 3: Calculate the change in x (depth): $70 - 15 = 55$.
- Step 4: Calculate the slope: $-9 / 55 \approx -0.1636$.
- Step 5: Compare this estimated slope to the answer choices: -0.1636 is closest to -0.17 .
- Step 6: This matches choice A.

Probability and conditional probability

2 questions · Easy 0 / Medium 0 / Hard 2

Probability and conditional probability — Q1 [Hard] [Grid-In] (ID: 89ff6a0a)

Question ID: 89ff6a0a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

For 100 buttons, the table summarizes the distribution by group and diameter.

Group	Diameter (millimeters)		
	Less than 20	20 to 30	Greater than 30
Group 1	12	8	3
Group 2	0	17	18
Group 3	10	32	0

One of these buttons will be selected at random. What is the probability of selecting a button with a diameter that is less than or equal to 30 millimeters, given that it is **not** in group 2? (Express your answer as a decimal or fraction, not as a percent.)

Correct Answer: 62/65 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Identify the condition ('given that it is NOT in group 2'): this means we only look at Group 1 and Group 3 rows, completely ignoring Group 2.
- Step 2: Calculate the total number of buttons in Group 1: $12 + 8 + 3 = 23$.
- Step 3: Calculate the total number of buttons in Group 3: $10 + 32 + 0 = 42$.
- Step 4: Calculate the combined total for 'not in group 2': $23 + 42 = 65$. This is our new denominator.
- Step 5: Now find how many of these 65 buttons have a diameter ≤ 30 mm — this means the 'Less than 20' AND '20 to 30' columns (both are ≤ 30 mm), but NOT the 'Greater than 30' column.
- Step 6: From Group 1: 'Less than 20' = 12, '20 to 30' = 8. Sum: $12 + 8 = 20$.
- Step 7: From Group 3: 'Less than 20' = 10, '20 to 30' = 32. Sum: $10 + 32 = 42$.
- Step 8: Combine these: $20 + 42 = 62$. This is our numerator.
- Step 9: The probability is $62/65$ (this fraction cannot be simplified further, since $62 = 2 \times 31$ and $65 = 5 \times 13$ share no common factors).
- Step 10: This is a grid-in question — enter $62/65$ (or its decimal equivalent, approximately 0.954) as your final answer.

Probability and conditional probability — Q2 [Hard] (ID: f496d2c0)

Question ID: f496d2c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

The table shows the distribution of rooms in a certain facility by seating capacity.

Seating capacity	Proportion
Less than 18 seats	26%
18–40 seats	21%
41–65 seats	29%
Greater than 65 seats	24%

If a room in this facility is selected at random, which of the following is closest to the probability of selecting a room that has a seating capacity greater than 65 seats, given that the room has a seating capacity of at least 18 seats?

Answer

- A. 0.24
- B. 0.32
- C. 0.50
- D. 0.92

Correct Answer: B

Step-by-Step Solution:

- Step 1: Identify the given condition: 'seating capacity of at least 18 seats' — this includes the '18–40 seats,' '41–65 seats,' and 'greater than 65 seats' categories (everything except 'less than 18 seats').
- Step 2: Add up the percentages for these three categories to find the new total: $21\% + 29\% + 24\% = 74\%$.
- Step 3: Identify what we're looking for within this reduced group: rooms with 'greater than 65 seats,' which is 24%.
- Step 4: Calculate the conditional probability by dividing the specific percentage by the new total: $24\% / 74\% \approx 0.3243$.
- Step 5: Rounding to match the answer choices, this is closest to 0.32.
- Step 6: This matches choice B.

Inference from sample statistics and margin of error

3 questions · Easy 0 / Medium 2 / Hard 1

Inference from sample statistics and margin of error — Q1 [Medium] (ID: 3e1bd4e2)

Question ID: 3e1bd4e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Medium

Question

Researchers studied a sample of a species of fish from a certain location. The researchers found that the mean length of fish in the sample was 16.59 millimeters. Based on the sample mean and margin of error, the researchers estimated that the mean length of all fish of this species from this location was between 15.86 and 17.32 millimeters. What is the margin of error, in millimeters?

Answer

- A. 0.73
- B. 2.19
- C. 15.86
- D. 16.59

Correct Answer: A

Step-by-Step Solution:

- Step 1: Identify the sample mean (the central estimate): 16.59 millimeters.
- Step 2: Identify one endpoint of the confidence interval — let's use the upper bound: 17.32 millimeters.
- Step 3: Calculate the margin of error by finding the difference between the upper bound and the sample mean: $17.32 - 16.59 = 0.73$.
- Step 4: Double-check using the lower bound: sample mean – margin of error should equal the lower bound: $16.59 - 0.73 = 15.86$, which matches the given lower bound exactly, confirming the calculation.
- Step 5: This matches choice A.

Inference from sample statistics and margin of error — Q2 [Medium] (ID: 2bab1399)

Question ID: 2bab1399

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Medium

Question

A researcher is designing a study to investigate the average number of hours participants in a training course spend on a particular activity per day. The researcher will report an estimated average number of hours participants in the training course spend on this activity per day with an associated margin of error. The researcher is considering using a random sample of either 115 or 230 participants from the training course. Which of the following would be the most likely effect of using the larger random sample compared to the smaller random sample?

Answer

- A. The reported margin of error would be lower.
- B. The reported margin of error would be higher.
- C. The reported average number of hours would be lower.
- D. The reported average number of hours would be higher.

Correct Answer: A

Step-by-Step Solution:

Step 1: Compare the two sample sizes being considered: 115 participants versus 230 participants.

Step 2: Recognize that 230 is the LARGER of the two sample sizes (twice as large as 115).

Step 3: Apply the general statistical principle: larger sample sizes lead to more precise estimates, which means a SMALLER (lower) margin of error.

Step 4: Since the question asks about the effect of using the LARGER sample (230) compared to the smaller one (115), the expected effect is a LOWER margin of error.

Step 5: This matches choice A.

Inference from sample statistics and margin of error — Q3 [Hard] (ID: da978ee6)

Question ID: da978ee6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard

Question

A researcher surveyed a random sample of 1,020 people from a certain city to estimate the percentage of people who support a recently announced proposal. From the survey, the researcher estimated that 86% of people from the city support this proposal, with an associated margin of error of 2.13%. The researcher repeated the survey with a random sample of people from the city that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

Answer

- A. The margin of error is between 2.13% and 3.13%.
- B. The margin of error is between 4.13% and 5.13%.
- C. The margin of error is greater than 5.13%.
- D. The margin of error is less than 2.13%.

Correct Answer: D

Step-by-Step Solution:

- Step 1: Identify the relationship between sample size and margin of error: margin of error is inversely proportional to the square root of the sample size.
- Step 2: Identify the change in sample size: the new sample is DOUBLE the original ($2 \times 1,020 = 2,040$ people).
- Step 3: Calculate the factor by which the margin of error changes: since sample size increased by a factor of 2, margin of error decreases by a factor of $\sqrt{2} \approx 1.414$.
- Step 4: Calculate the new (approximate) margin of error: $2.13\% \div 1.414 \approx 1.506\%$.
- Step 5: Compare this new margin of error (approximately 1.51%) to the original (2.13%) and to the answer choices — 1.51% is LESS than 2.13%, matching the description in choice D.
- Step 6: This matches choice D.

Geometry and Trigonometry

Math · 51 questions · Εκφώνηση / Correct Answer / Step-by-Step Solution

Area and volume

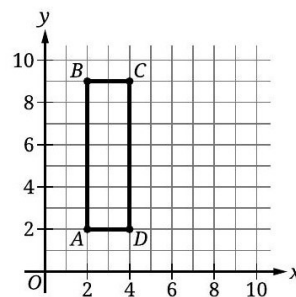
16 questions · Easy 5 / Medium 3 / Hard 8

Area and volume — Q1 [Easy] (ID: 124bc42b)

Question ID: 124bc42b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Rectangle $ABCD$ shown has a length of 7 units and a width of 2 units. What is the area, in square units, of rectangle $ABCD$?

Answer

- A. 5
- B. 9
- C. 14
- D. 28

Correct Answer: C

Step-by-Step Solution:

- Step 1: Identify the given dimensions: length = 7 units, width = 2 units.
- Step 2: Apply the area formula for a rectangle: Area = length $\times$ width.
- Step 3: Substitute the values: Area = $7 \times 2 = 14$.
- Step 4: This matches choice C.

Area and volume — Q2 [Easy] (ID: c8ab85ad)

Question ID: c8ab85ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question
Which expression represents the area, in square inches, of a square with a side length of 35 inches?

- Answer**
- A. $\sqrt{35}$
 - B. $35 + 35$
 - C. $(4)(35)$
 - D. 35^2

Correct Answer: D

Step-by-Step Solution:

Step 1: Recall the area formula for a square: $\text{Area} = (\text{side length})^2$.

Step 2: Substitute the given side length: $\text{Area} = 35^2$.

Step 3: This matches choice D exactly — no further calculation is needed since the question asks for the EXPRESSION, not a numeric value.

Area and volume — Q3 [Easy] (ID: f3f06c3a)

Question ID: f3f06c3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy



Note: Figure not drawn to scale.

- In the rectangle shown, $x = 7$ centimeters and $y = 5$ centimeters. What is the area, in square centimeters, of the rectangle?
- Answer**
- A. 7
 - B. 12
 - C. 35
 - D. 40

Correct Answer: C

Step-by-Step Solution:

Step 1: Identify the given dimensions: $x = 7$ centimeters, $y = 5$ centimeters.

Step 2: Apply the area formula for a rectangle: $\text{Area} = \text{length} \times \text{width} = x \times y$.

Step 3: Substitute the values: $\text{Area} = 7 \times 5 = 35$.

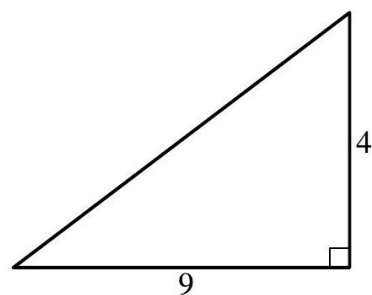
Step 4: This matches choice C.

Area and volume — Q4 [Easy] (ID: cab8f907)

Question ID: cab8f907

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Note: Figure not drawn to scale.

The figure shows the lengths, in units, of two sides of a triangle. What is the area, in square units, of the triangle?

Answer

- A. 18
- B. 13
- C. 9
- D. 4

Correct Answer: A

Step-by-Step Solution:

Step 1: Identify the base and height from the figure: the right-angle marker confirms the sides labeled 9 and 4 are perpendicular to each other, making them the base and height.

Step 2: Apply the triangle area formula: $\text{Area} = (1/2) \times \text{base} \times \text{height}$.

Step 3: Substitute the values: $\text{Area} = (1/2) \times 9 \times 4$.

Step 4: Calculate: $9 \times 4 = 36$, then $(1/2) \times 36 = 18$.

Step 5: This matches choice A.

Area and volume — Q5 [Easy] (ID: 17049054)

Question ID: 17049054

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

The length of a side of square X is 9 centimeters. The area of rectangle Y is 32 square centimeters. What is the total area, in square centimeters, of square X and rectangle Y?

Answer

- A. 145
- B. 113
- C. 82
- D. 81

Correct Answer: B

Step-by-Step Solution:

- Step 1: Calculate the area of square X using the given side length: $\text{Area} = \text{side}^2 = 9^2 = 81$ square centimeters.
- Step 2: Note the area of rectangle Y, which is given directly in the problem: 32 square centimeters.
- Step 3: Add the two areas together to find the total: $81 + 32 = 113$.
- Step 4: This matches choice B.

Area and volume — Q6 [Medium] [Grid-In] (ID: 25cc5327)

Question ID: 25cc5327

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The base of a right rectangular pyramid has a length of 27 centimeters and a width of 2 centimeters. The height of this pyramid is 7 centimeters. What is the volume, in cubic centimeters, of this pyramid?

Correct Answer: 126 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Calculate the base area of the rectangular pyramid: $\text{base area} = \text{length} \times \text{width} = 27 \times 2 = 54$ square centimeters.
- Step 2: Apply the pyramid volume formula: $\text{Volume} = (1/3) \times \text{base area} \times \text{height}$.
- Step 3: Substitute the values: $\text{Volume} = (1/3) \times 54 \times 7$.
- Step 4: Calculate $54 \times 7 = 378$, then $(1/3) \times 378 = 126$.
- Step 5: This is a grid-in question — enter 126 as your final answer.

Area and volume — Q7 [Medium] (ID: 42155bac)

Question ID: 42155bac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The area of one of the bases of a right circular cylinder is 81 square centimeters, and the volume of the cylinder is 2,187 cubic centimeters. What is the height, in centimeters, of the cylinder?

Answer

- A. 27
- B. 2,025
- C. 2,106
- D. 177,147

Correct Answer: A

Step-by-Step Solution:

- Step 1: Recall the cylinder volume formula: $\text{Volume} = \text{base area} \times \text{height}$.
- Step 2: Rearrange this formula to solve for height: $\text{height} = \text{Volume} \div \text{base area}$.
- Step 3: Substitute the given values: $\text{height} = 2,187 \div 81$.
- Step 4: Calculate: $2,187 \div 81 = 27$.
- Step 5: This matches choice A.

Area and volume — Q8 [Medium] [Grid-In] (ID: 1b687ae3)

Question ID: 1b687ae3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The area of a triangle is 36 square units. If the height of this triangle is 2 units, what is the length, in units, of the base of this triangle?

Correct Answer: 36 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Recall the triangle area formula: $\text{Area} = (1/2) \times \text{base} \times \text{height}$.
- Step 2: Rearrange this formula to solve for the base: $\text{base} = (2 \times \text{Area}) \div \text{height}$.
- Step 3: Substitute the given values: $\text{base} = (2 \times 36) \div 2$.
- Step 4: Calculate the numerator first: $2 \times 36 = 72$.
- Step 5: Divide: $72 \div 2 = 36$.
- Step 6: This is a grid-in question — enter 36 as your final answer.

Area and volume — Q9 [Hard] (ID: 1fe41c6b)

Question ID: 1fe41c6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	2,520	n

What is the value of n ?

Answer

- A. 2,100
- B. 1,680
- C. 840
- D. 420

Correct Answer: D

Step-by-Step Solution:

Step 1: Calculate the area ratio between the two similar rectangles: Rectangle B's area $\div$ Rectangle A's area = $2,520 \div 630 = 4$.

Step 2: Since area scales as the SQUARE of the linear scale factor, find the actual scale factor by taking the square root of the area ratio: $\sqrt{4} = 2$.

Step 3: Since perimeter scales LINEARLY (directly proportional to the scale factor, not squared), multiply Rectangle A's perimeter by this same scale factor: $n = 210 \times 2$.

Step 4: Calculate: $210 \times 2 = 420$.

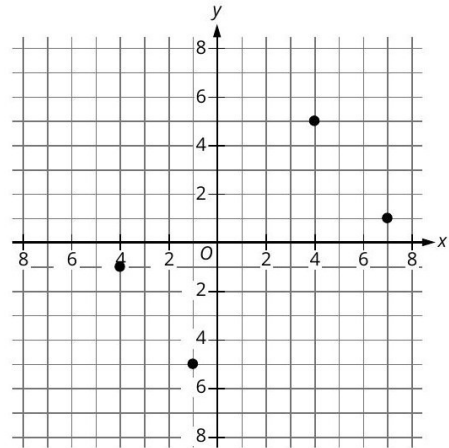
Step 5: This matches choice D.

Area and volume — Q10 [Hard] [Grid-In] (ID: 30aa51b7)

Question ID: 30aa51b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

Correct Answer: 50 (Student-Produced Response — no answer choices given)

- Step-by-Step Solution:**
- Step 1: Read the four vertex coordinates from the graph: approximately $(-4, -1)$, $(4, 5)$, $(7, 1)$, and $(-1, -5)$.
- Step 2: Calculate the length of one side using the distance formula between two adjacent vertices, $(-4, -1)$ and $(4, 5)$:
distance = $\sqrt{[(4 - (-4))]^2 + [(5 - (-1))]^2} = \sqrt{[(8)]^2 + [(6)]^2} = \sqrt{64 + 36} = \sqrt{100} = 10$.
- Step 3: Calculate the length of the adjacent side between $(4, 5)$ and $(7, 1)$: distance = $\sqrt{[(7 - 4)]^2 + [(1 - 5)]^2} = \sqrt{[(3)]^2 + [(-4)]^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.
- Step 4: Verify these two sides are perpendicular (confirming it's truly a rectangle) by checking their slopes multiply to -1 : slope of first side = $6/8 = 3/4$; slope of second side = $-4/3$. Product: $(3/4) \times (-4/3) = -1$. Confirmed perpendicular.
- Step 5: Calculate the area by multiplying the two side lengths: Area = $10 \times 5 = 50$.
- Step 6: This is a grid-in question — enter 50 as your final answer.

Area and volume — Q11 [Hard] [Grid-In] (ID: 179edb12)

Question ID: 179edb12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard ■■■■■

Question

A right circular cone has a height of 22 centimeters (cm) and a base with a diameter of 18 cm. The volume of this cone is $n\pi$ cm³. What is the value of n ?

Correct Answer: 594 (Student-Produced Response — no answer choices given)**Step-by-Step Solution:**

Step 1: Convert the given diameter to radius: radius = diameter $\div$ 2 = $18 \div 2 = 9$ cm.

Step 2: Recall the cone volume formula: Volume = $(1/3) \times \pi \times r^2 \times \text{height}$.

Step 3: Substitute the values: Volume = $(1/3) \times \pi \times 9^2 \times 22$.

Step 4: Calculate $9^2 = 81$.

Step 5: Multiply: $81 \times 22 = 1,782$.

Step 6: Multiply by $(1/3)$: $(1/3) \times 1,782 = 594$.

Step 7: The volume is 594π cubic centimeters, so comparing to the given form $n\pi$, we get $n = 594$.

Step 8: This is a grid-in question — enter 594 as your final answer.

Area and volume — Q12 [Hard] (ID: e6c557f9)

Question ID: e6c557f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard ■■■■■

Question

The edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n ?

Answer

- A. $\frac{9}{7,744}$
B. $\frac{27}{3,872}$
C. $\frac{3}{88}$
D. $\frac{9}{44}$

Correct Answer: A**Step-by-Step Solution:**

Step 1: Identify the given linear (edge length) ratio: cube Y's edge is $3/88$ of cube X's edge.

Step 2: Recall that surface area scales as the SQUARE of the linear scale factor for similar 3D shapes.

Step 3: Square the given ratio: $n = (3/88)^2 = (3^2)/(88^2) = 9/7,744$.

Step 4: This matches choice A.

Area and volume — Q13 [Hard] [Grid-In] (ID: 73e1b793)

Question ID: 73e1b793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is 972 cubic centimeters (cm^3), and the volume of prism Q is 36 cm^3 . One edge of prism P has a length of 12 cm. What is the length, in cm, of the corresponding edge of prism Q?

Correct Answer: 4 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

Step 1: Calculate the volume ratio between the two similar prisms: prism P's volume $\div$ prism Q's volume = $972 \div 36 = 27$.

Step 2: Since volume scales as the CUBE of the linear scale factor, find the actual scale factor by taking the cube root of the volume ratio: the cube root of 27 is 3 (since $3 \times 3 \times 3 = 27$).

Step 3: This means every linear dimension of prism P is 3 times the corresponding dimension of prism Q.

Step 4: Since the given edge of prism P is 12 cm, find the corresponding edge of prism Q by dividing by the scale factor: $12 \div 3 = 4$.

Step 5: This is a grid-in question — enter 4 as your final answer.

Area and volume — Q14 [Hard] (ID: 9fdc4eaa)

Question ID: 9fdc4eaa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right square pyramid has a total surface area of 70,560 square inches, and the combined surface area of the four lateral faces of this pyramid is 38,160 square inches. What is the height, in inches, of this pyramid?

Answer

- A. 56
- B. 90
- C. 106
- D. 180

Correct Answer: A

Step-by-Step Solution:

Step 1: Find the base area by subtracting the lateral surface area from the total surface area: base area = $70,560 - 38,160 = 32,400$ square inches.

Step 2: Since the base is a square, find its side length by taking the square root: side length = $\sqrt{32,400} = 180$ inches.

Step 3: Use the lateral surface area formula for a square pyramid (4 triangular faces, each with area $(1/2) \times$ base side $\times$ slant height, combined = $2 \times$ base side $\times$ slant height): $38,160 = 2 \times 180 \times$ slant height.

Step 4: Solve for slant height: slant height = $38,160 \div 360 = 106$ inches.

Step 5: Set up the Pythagorean theorem relating the pyramid's height, half the base side length, and the slant height (these three form a right triangle): $\text{height}^2 + (\text{base side}/2)^2 = \text{slant height}^2$.

Step 6: Substitute the values: $\text{height}^2 + 90^2 = 106^2$.

Step 7: Calculate: $\text{height}^2 + 8,100 = 11,236$.

Step 8: Solve for height^2 : $\text{height}^2 = 11,236 - 8,100 = 3,136$.

Step 9: Take the square root: height = $\sqrt{3,136} = 56$.

Step 10: This matches choice A.

Area and volume — Q15 [Hard] [Grid-In] (ID: f6ca90cc)

Question ID: f6ca90cc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard ■■■■■■■■■■

Question

A right prism has a square base with side lengths of 10 inches. The height of the prism is 58 inches. What is the volume, in cubic inches, of the prism?

Correct Answer: 5800 (Student-Produced Response — no answer choices given)**Step-by-Step Solution:**

Step 1: Calculate the area of the square base: $\text{base area} = \text{side}^2 = 10^2 = 100$ square inches.

Step 2: Apply the prism volume formula: $\text{Volume} = \text{base area} \times \text{height}$.

Step 3: Substitute the values: $\text{Volume} = 100 \times 58$.

Step 4: Calculate: $100 \times 58 = 5,800$.

Step 5: This is a grid-in question — enter 5,800 as your final answer.

Area and volume — Q16 [Hard] [Grid-In] (ID: 4757123b)

Question ID: 4757123b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard ■■■■■■■■■■

Question

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is 41 times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

Correct Answer: 1681 (Student-Produced Response — no answer choices given)**Step-by-Step Solution:**

Step 1: Identify the given linear scale factor: each side of trapezoid A is 41 times the corresponding side of trapezoid B.

Step 2: Recall that area scales as the SQUARE of the linear scale factor for similar figures.

Step 3: Square the given scale factor: $41^2 = 41 \times 41 = 1,681$.

Step 4: This means trapezoid A's area is 1,681 times as large as trapezoid B's area.

Step 5: This is a grid-in question — enter 1,681 as your final answer.

Lines, angles, and triangles

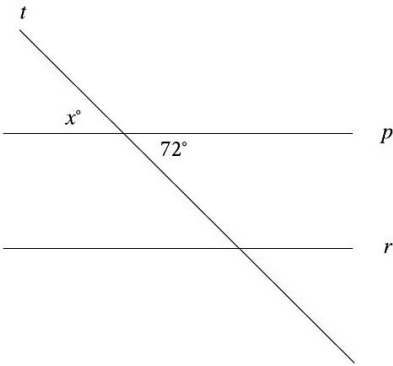
14 questions · Easy 2 / Medium 6 / Hard 6

Lines, angles, and triangles — Q1 [Easy] (ID: 4a141e77)

Question ID: 4a141e77

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line p is parallel to line r , and line t intersects both lines. What is the value of x ?

Answer

- A. 36
- B. 72
- C. 180
- D. 252

Correct Answer: B

Step-by-Step Solution:

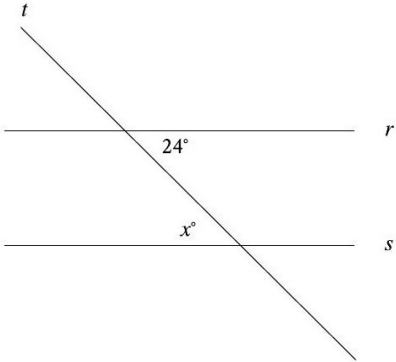
- Step 1: Identify the two labeled angles' positions: x° and 72° are both located at the single intersection point where line t crosses line p .
- Step 2: Determine their relationship: they are diagonally opposite each other at this intersection, making them vertical angles.
- Step 3: Apply the vertical angles theorem: vertical angles are always equal, regardless of any other properties in the figure (like whether p is parallel to r).
- Step 4: Since 72° is given, $x = 72$.
- Step 5: This matches choice B.

Lines, angles, and triangles — Q2 [Easy] (ID: d21270da)

Question ID: d21270da

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line t intersects lines r and s . Which additional piece of information is sufficient to prove that lines r and s are parallel?

Answer

- A. $x < 24$
- B. $x = 24$
- C. $x > 24$
- D. $x - 24 = 360$

Correct Answer: B

Step-by-Step Solution:

- Step 1: Identify the positions of the two labeled angles: 24° is at the intersection of t and r ; x° is at the intersection of t and s .
- Step 2: Compare their relative positions: both angles sit in the same location relative to their own intersection (same side of line t , same side of the horizontal line) — this makes them corresponding angles.
- Step 3: Recall the Corresponding Angles Postulate: two lines cut by a transversal are parallel IF AND ONLY IF corresponding angles are equal.
- Step 4: Apply this to find the necessary condition: for r and s to be parallel, x must equal 24.
- Step 5: This matches choice B.

Lines, angles, and triangles — Q3 [Medium] (ID: e89f63bf)

Question ID: e89f63bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

The sum of the measures of two angles in a triangle is 64° . What is the measure of the third angle?

Answer

- A. 26°
- B. 52°
- C. 116°
- D. 128°

Correct Answer: C

Step-by-Step Solution:

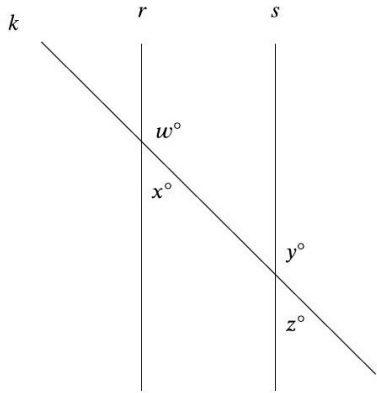
- Step 1: Recall the triangle angle sum theorem: the three interior angles of any triangle always add up to 180° .
- Step 2: The problem states two of the angles sum to 64° .
- Step 3: Find the third angle by subtracting this sum from 180° : $180 - 64 = 116$.
- Step 4: This matches choice C.

Lines, angles, and triangles — Q4 [Medium] (ID: ecc98c87)

Question ID: ecc98c87

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, line k intersects lines r and s . If $w = 147$, which additional piece of information is sufficient to prove that lines r and s are parallel?

Answer

- A. $x = 33$
- B. $y = 147$
- C. $w + y = 180$
- D. $y + z = 180$

Correct Answer: B

Step-by-Step Solution:

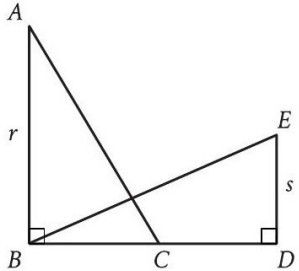
- Step 1: Identify the given angle: $w = 147^\circ$ at the r intersection.
- Step 2: Determine which angle at the s intersection occupies the SAME relative position as w — comparing the figure, this is y° (both are in the upper-right region relative to their intersection).
- Step 3: Recall the Corresponding Angles Postulate: for two lines cut by a transversal to be parallel, corresponding angles must be equal.
- Step 4: Apply this: since $w = 147$, for $r \parallel s$, we need $y = 147$ as well.
- Step 5: This matches choice B.

Lines, angles, and triangles — Q5 [Medium] (ID: 697b3954)

Question ID: 697b3954

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, $\triangle ABC$ is congruent to $\triangle BDE$. If $r = 54$ and $s = 26$, what is the length of $\overline{CD}$?

Answer

- A. 26
- B. 27
- C. 28
- D. 54

Correct Answer: C

Step-by-Step Solution:

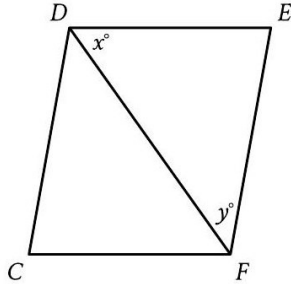
- Step 1: Write out the vertex correspondence from the congruence statement: triangle $ABC \cong$ triangle BDE means $A \leftrightarrow B$, $B \leftrightarrow D$, $C \leftrightarrow E$.
- Step 2: Match up corresponding sides: side AB (in triangle ABC) corresponds to side BD (in triangle BDE); side BC corresponds to side DE .
- Step 3: Use the given value $r = 54$ for side AB (the vertical leg in the figure): since AB corresponds to BD , this means $BD = 54$.
- Step 4: Use the given value $s = 26$ for side DE (the vertical leg on the right side of the figure): since BC corresponds to DE , this means $BC = 26$.
- Step 5: Notice from the figure that B , C , and D are all on the same horizontal line, with C between B and D — so $BD = BC + CD$.
- Step 6: Substitute the known values: $54 = 26 + CD$.
- Step 7: Solve for CD : $CD = 54 - 26 = 28$.
- Step 8: This matches choice C.

Lines, angles, and triangles — Q6 [Medium] (ID: ffc39dc5)

Question ID: ffc39dc5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, $CD = EF$ and $FC = DE$. If $x = 65$ and $y = 30$, what is the measure of $\angle CDE$?

Answer

- A. 30°
- B. 65°
- C. 95°
- D. 130°

Correct Answer: C

Step-by-Step Solution:

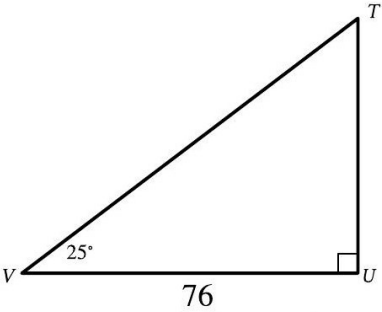
- Step 1: Identify the given side equalities: $CD = EF$ and $FC = DE$, plus the shared diagonal DF (common to both triangles CDF and EFD).
- Step 2: Recognize this establishes SSS congruence between triangle CDF and triangle EFD , with correspondence $C \leftrightarrow E$, $D \leftrightarrow F$, $F \leftrightarrow D$.
- Step 3: From this congruence, corresponding angles are equal: angle CDF (at vertex D in triangle CDF) corresponds to angle EFD (at vertex F in triangle EFD), which is given as $y = 30^\circ$. So angle $CDF = 30^\circ$.
- Step 4: Identify that the full angle CDE (the angle the question asks about) is made up of two parts: angle CDF (just found to be 30°) plus angle FDE (which is given as $x = 65^\circ$, the angle between DF and DE at vertex D).
- Step 5: Add these two parts together: angle $CDE = \text{angle } CDF + \text{angle } FDE = 30 + 65 = 95$.
- Step 6: This matches choice C.

Lines, angles, and triangles — Q7 [Medium] (ID: 8364487a)

Question ID: 8364487a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

Triangle TUV is dilated by a scale factor of $\frac{1}{2}$ to obtain triangle $T'U'V'$ (not shown), where T corresponds to T' and U corresponds to U' . What is the measure, in degrees, of angle V' ?

Answer

- A. 25
- B. 38
- C. 50
- D. 76

Correct Answer: A

Step-by-Step Solution:

Step 1: Recall the fundamental property of dilation: it changes side lengths (by the scale factor) but NEVER changes angle measures.

Step 2: Identify the original angle V in triangle TUV : 25° (given in the figure).

Step 3: Since angle V' in the dilated triangle $T'U'V'$ corresponds to angle V , and dilation preserves angles, angle $V' = \text{angle } V = 25^\circ$.

Step 4: This matches choice A.

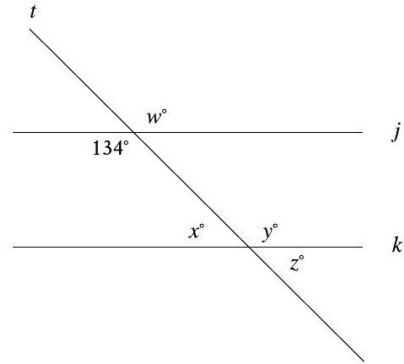
(Note: the scale factor of $\frac{1}{2}$ and the side length 76 given in the problem are both irrelevant distractors for this specific question — they would matter if the question asked for a SIDE LENGTH in the dilated triangle, but for an ANGLE measure, only the original angle value matters.)

Lines, angles, and triangles — Q8 [Medium] (ID: 0013adbd)

Question ID: 0013adbd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure, line t intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

Answer

- A. $x = 46$
- B. $y = 46$
- C. $w = 134$
- D. $z = 134$

Correct Answer: A

Step-by-Step Solution:

- Step 1: Identify the positions of the two labeled angles: 134° is at the j intersection; x° is at the k intersection.
- Step 2: Determine their relationship: both are on the same side of transversal t , and both face into the region BETWEEN lines j and k — this makes them same-side interior (co-interior) angles.
- Step 3: Recall the Same-Side Interior Angles Theorem: for two lines cut by a transversal to be parallel, same-side interior angles must be SUPPLEMENTARY (sum to 180°).
- Step 4: Set up the equation: $134 + x = 180$.
- Step 5: Solve for x : $x = 180 - 134 = 46$.
- Step 6: This matches choice A.

Lines, angles, and triangles — Q9 [Hard] (ID: 1dc7e423)

Question ID: 1dc7e423

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle ABC , the measure of angle A is 52° and $AC = 30$. In triangle PQR , the measure of angle P is 52° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

Answer

- A. $AB = 50$ and $PQ = 50$.
- B. $AB = 50$ and $QR = 200$.
- C. The measures of angle B and angle R are 32° and 96° , respectively.
- D. The measures of angle B and angle Q are 52° and 32° , respectively.

Correct Answer: B

Step-by-Step Solution:

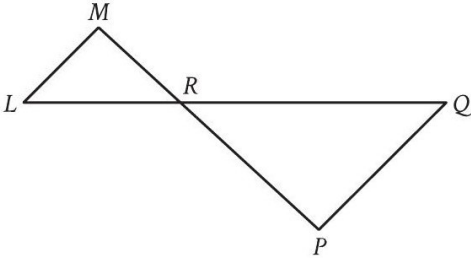
- Step 1: Note the given information: angle $B = \text{angle } G = 51^\circ$, $AB = 6$, $FG = 18$ (so $AB/FG = 1/3$).
- Step 2: Evaluate option I (angle $A = \text{angle } F$): combined with the given angle $B = \text{angle } G$, this provides two pairs of equal angles, which is sufficient for AA similarity — I is SUFFICIENT.
- Step 3: Evaluate option II (angle $C = \text{angle } H$): similarly, combined with angle $B = \text{angle } G$, this gives two pairs of equal angles, sufficient for AA similarity — II is SUFFICIENT.
- Step 4: Evaluate option III ($FH = 3 \times AC$): check whether angle B/G is INCLUDED between the sides being compared (AB & FG , and now AC & FH). Side AC is OPPOSITE angle B (not adjacent to it), so angle B is NOT the included angle between AB and AC . This means the given ratio conditions ($AB/FG = 1/3$ and implied $AC/FH = 1/3$) don't set up a valid SAS criterion — this is an ambiguous SSA situation, which does NOT guarantee similarity — III is NOT SUFFICIENT.
- Step 5: Combining these results: I is sufficient, II is sufficient, but III is not — matching choice B.

Lines, angles, and triangles — Q10 [Hard] (ID: b5d62bba)

Question ID: b5d62bba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

Answer

- A. $\frac{576}{49}$
B. $\frac{144}{7}$
C. 63
D. $\frac{441}{4}$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Recognize that since $LM \parallel PQ$, triangles LMR and QPR are similar (by AA: vertical angles at R are equal, and alternate interior angles formed by the parallel lines and the transversals LQ, MP are equal).
- Step 2: Identify the correct corresponding sides: MR (in triangle LMR) corresponds to RP (in triangle QPR), since these are the sides connecting to point R from each parallel segment.
- Step 3: Calculate the scale factor (ratio of corresponding sides): $MR/RP = 8/14 = 4/7$.
- Step 4: Since triangle PQR is LARGER ($RP=14 > MR=8$), we need the ratio of PQR's area to LMR's area, which uses the INVERSE scale factor squared: $(RP/MR)^2 = (14/8)^2 = (7/4)^2 = 49/16$.
- Step 5: Calculate the area of triangle PQR: $\text{Area}(PQR) = \text{Area}(LMR) \times (49/16) = 36 \times 49/16$.
- Step 6: Calculate: $36 \times 49 = 1,764$. Then $1,764 \div 16 = 110.25$, which as a fraction is $441/4$.
- Step 7: This matches choice D.

Lines, angles, and triangles — Q11 [Hard] (ID: 7ce2e728)

Question ID: 7ce2e728

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

For triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6, and the length of $\overline{FG}$ is 18. Which additional piece(s) of information is(are) sufficient to prove that triangle ABC is similar to triangle FGH ?

- I. The measure of angle A is equal to the measure of angle F .
- II. The measure of angle C is equal to the measure of angle H .
- III. The length of $\overline{FH}$ is 3 times the length of $\overline{AC}$.

Answer

- A. I is sufficient, II is sufficient, and III is sufficient.
- B. I is sufficient and II is sufficient, but III is not.
- C. II is sufficient and III is sufficient, but I is not.
- D. III is sufficient, but I is not and II is not.

Correct Answer: B

Step-by-Step Solution:

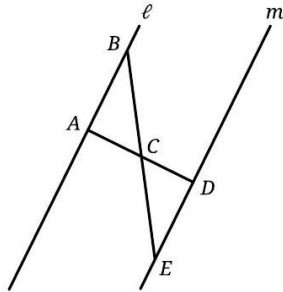
- Step 1: Note the given information: angle $B = \text{angle } G = 51^\circ$, $AB = 6$, $FG = 18$ (giving a ratio $AB/FG = 1/3$).
- Step 2: Evaluate option I (angle $A = \text{angle } F$): combined with the already-given angle $B = \text{angle } G$, this establishes TWO pairs of equal angles — sufficient for AA similarity.
- Step 3: Evaluate option II (angle $C = \text{angle } H$): similarly combines with angle $B = \text{angle } G$ for a second angle pair — also sufficient for AA similarity.
- Step 4: Evaluate option III ($FH = 3 \times AC$): check whether angle B (the given equal angle) is INCLUDED between the sides being compared. Side AC is OPPOSITE angle B in triangle ABC (not adjacent to it) — so this is an SSA setup (side-side-angle where the angle isn't between the two sides), which does NOT reliably prove similarity.
- Step 5: Conclude: I is sufficient, II is sufficient, but III is not — matching choice B.

Lines, angles, and triangles — Q12 [Hard] (ID: 7a11ceea)

Question ID: 7a11ceea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, segment AD and segment BE intersect at point C . Segment AD is perpendicular to line ℓ at point A . Which of the following additional pieces of information is(are) sufficient to determine that triangle ABC is congruent to triangle DEC ?

- I. Segment AD is perpendicular to line m at point D .
- II. $AC = 13$ and $DC = 13$

Answer

- A. I is sufficient by itself, but II is not.
- B. II is sufficient by itself, but I is not.
- C. I is sufficient by itself, and II is sufficient by itself.
- D. Neither I nor II is sufficient by itself.

Correct Answer: D

Step-by-Step Solution:

Step 1: Identify what's already established from the figure regardless of any additional information: angle $BAC = 90^\circ$ (since $AD \perp \ell$ at A is given), and angle $ACB = \text{angle } DCE$ (vertical angles at point C , always true).

Step 2: Evaluate statement I alone ($AD \perp m$ at D): this adds angle $EDC = 90^\circ$. Combined with what we already know, we now have TWO pairs of equal angles (angle $BAC = \text{angle } EDC = 90^\circ$, and angle $ACB = \text{angle } DCE$). This is AA, which proves the triangles are SIMILAR, but similarity alone does NOT prove congruence (the triangles could be different sizes) — I alone is NOT sufficient for congruence.

Step 3: Evaluate statement II alone ($AC = DC = 13$): this adds one side equality. Combined with the always-true vertical angle equality, we have only ONE side pair and ONE angle pair equal — this is not a complete congruence criterion (we'd need either a second side for SAS or a second angle for ASA) — II alone is NOT sufficient.

Step 4: Since neither I nor II alone provides a complete congruence criterion, the correct answer is that neither is sufficient by itself.

Step 5: This matches choice D.

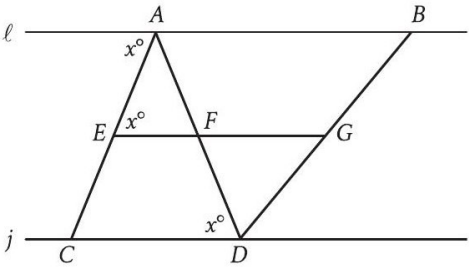
Lines, angles, and triangles — Q13 [Hard] (ID: 858809e7)

Question ID: 858809e7



Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{EG}$ and $\overline{AD}$ intersect at point F , and line ℓ is parallel to line j . If $EF = 0.5CD$ and $FG = 2.25CD$, which of the following must be true?

- I. $AB = 2.25EF$
- II. $FG = 0.5AB$
- III. $\frac{EF}{FG} = \frac{AF}{FD}$

Answer

- A. I only
- B. II only
- C. I and II
- D. II and III

Correct Answer: B

Step-by-Step Solution:

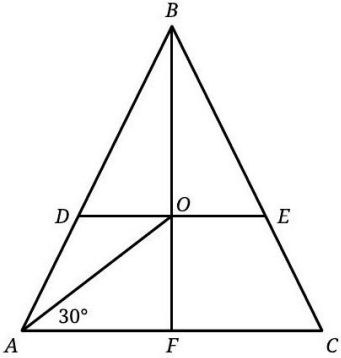
- Step 1: Confirm $EF \parallel CD$ using the matching x° angle markers at A and E (corresponding angles equal implies parallel lines).
- Step 2: Since $EF \parallel CD$, triangle AEF is similar to triangle ACD (by AA). This gives the proportion: $AF/AD = EF/CD$.
- Step 3: Substitute the given $EF = 0.5 \times CD$: $AF/AD = 0.5$. This means F is exactly the MIDPOINT of segment AD.
- Step 4: Since E, F, and G lie on the same straight horizontal line (given as segment EG), and EF is horizontal (parallel to CD, hence parallel to AB too, since AB and CD are both parts of the parallel lines ℓ and j), the continuation FG is ALSO horizontal, making $FG \parallel AB$.
- Step 5: Since $FG \parallel AB$, triangle DFG is similar to triangle DAB (by AA). This gives the proportion: $FG/AB = DF/DA$.
- Step 6: Since F is the midpoint of AD (from Step 3), $DF/DA = 0.5$ as well ($DF = FA$, both are half of AD).
- Step 7: Substitute into the proportion: $FG/AB = 0.5$, which means $FG = 0.5 \times AB$ — this confirms statement II is TRUE.
- Step 8: Check statement I: from Step 7, $AB = 2 \times FG = 2 \times 2.25CD = 4.5CD$. Compare to $2.25 \times EF = 2.25 \times 0.5CD = 1.125CD$. Since $4.5CD \neq 1.125CD$, statement I is FALSE.
- Step 9: Check statement III: $EF/FG = 0.5CD/2.25CD = 2/9$. But $AF/FD = 1$ (since F is the exact midpoint, AF and FD are equal). Since $2/9 \neq 1$, statement III is FALSE.
- Step 10: Only statement II is true, matching choice B.

Lines, angles, and triangles — Q14 [Hard] [Grid-In] (ID: 1dacfb94)

Question ID: 1dacfb94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In triangle ABC , $AB = BC$. Points D and E lie on line segments AB and BC , respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE , and the measure of $\angle OAC$ is 30° . Point F lies on line segment AC , and line segment BF passes through point O . If $AB = 81$ and $AO = BO$, what is the length of line segment BE ?

Correct Answer: 54 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Recognize that since $AB=BC$ (isosceles) and $DE\parallel AC$ with O as the midpoint of DE , the line BF (through B and O) is the **AXIS OF SYMMETRY** of the triangle — meaning F is the midpoint of AC , and BF is perpendicular to AC .
- Step 2: Set up coordinates for convenience: place F at the origin $(0,0)$, A at $(-40.5, 0)$, C at $(40.5, 0)$ — using half of AB 's eventual determined relationship — and B directly above F .
- Step 3: Using the given angle $OAC=30^\circ$ and the condition $AO=BO$, work through the triangle's angle relationships: since $BF\perp AC$ and bisects angle B , and using the isosceles triangle's base angles, algebraic substitution shows that angle $BAC = \text{angle } BCA = 60^\circ$, meaning triangle ABC is actually **EQUILATERAL** (since $AB=BC=81$ and the apex angle also works out to 60° , forcing $AC=81$ too).
- Step 4: With the equilateral triangle established, place coordinates: $A=(-40.5,0)$, $C=(40.5,0)$, $B=(0, 40.5\sqrt{3})$ (using the height of an equilateral triangle with side 81).
- Step 5: Find O 's exact position along BF (the y -axis) using the condition $AO=BO$: setting $O=(0,h)$, solve $\sqrt{(40.5^2+h^2)} = 40.5\sqrt{3}-h$, which gives $h = 13.5\sqrt{3}$ after algebraic simplification.
- Step 6: Find point E on segment BC where the horizontal line through O (at height $h=13.5\sqrt{3}$) intersects BC : parametrizing BC from $B(0,40.5\sqrt{3})$ to $C(40.5,0)$, solving for the parameter where $y=13.5\sqrt{3}$ gives a parameter value of $2/3$ (measuring from B toward C).
- Step 7: Calculate BE using this parameter: $BE = (2/3) \times BC = (2/3) \times 81 = 54$.
- Step 8: This is a grid-in question — enter 54 as your final answer.

Right triangles and trigonometry

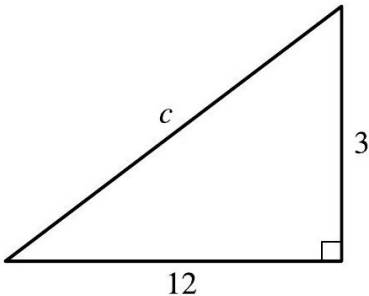
9 questions · Easy 1 / Medium 1 / Hard 7

Right triangles and trigonometry — Q1 [Easy] (ID: de9148c4)

Question ID: de9148c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Easy

Question



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the triangle shown?

Answer

- A. $12 \cdot 3 = c$
- B. $12 + 3 = c$
- C. $12^2 + 3^2 = c^2$
- D. $12^2 - 3^2 = c^2$

Correct Answer: C

Step-by-Step Solution:

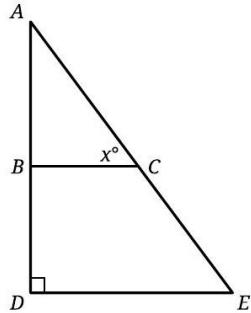
- Step 1: Identify the right angle's location in the figure: it's marked at the bottom-right vertex.
- Step 2: Identify the hypotenuse: it's the side OPPOSITE the right angle, which is side c (the slanted/diagonal side).
- Step 3: Identify the two legs: the horizontal side (12) and the vertical side (3).
- Step 4: Apply the Pythagorean theorem: $\text{leg}^2 + \text{leg}^2 = \text{hypotenuse}^2$, giving $12^2 + 3^2 = c^2$.
- Step 5: This matches choice C.

Right triangles and trigonometry — Q2 [Medium] (ID: 2757549e)

Question ID: 2757549e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the figure, $\overline{BC}$ is parallel to $\overline{DE}$. If the length of $\overline{DE}$ is 147 and the length of $\overline{AE}$ is 245, what is the value of $\tan x^\circ$?

Answer

- A. $\frac{147}{196}$
- B. $\frac{147}{245}$
- C. $\frac{196}{147}$
- D. $\frac{245}{147}$

Correct Answer: C

Step-by-Step Solution:

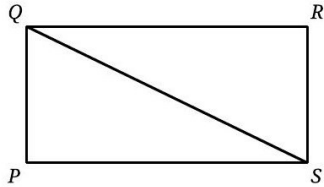
- Step 1: Recognize that since $BC \parallel DE$, the angle x° (angle BCA) corresponds to angle DEA (angle AED in the larger triangle) — corresponding angles are equal when lines are parallel.
- Step 2: Focus on the larger right triangle ADE (right angle at D), which has hypotenuse $AE = 245$ and leg $DE = 147$.
- Step 3: Find the missing leg AD using the Pythagorean theorem: $AD^2 = AE^2 - DE^2 = 245^2 - 147^2$.
- Step 4: Calculate: $245^2 = 60,025$, and $147^2 = 21,609$. So $AD^2 = 60,025 - 21,609 = 38,416$.
- Step 5: Take the square root: $AD = \sqrt{38,416} = 196$.
- Step 6: Set up the tangent ratio for angle AED (which equals x°): $\tan(x^\circ) = \text{opposite/adjacent} = AD/DE = 196/147$.
- Step 7: This matches choice C.

Right triangles and trigonometry — Q3 [Hard] [Grid-In] (ID: 2c3aefc9)

Question ID: 2c3aefc9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In rectangle $PQRS$, the length of line segment QS is $\sqrt{2,344}$ and the length of side PS is 8 more than the length of side RS . What is the length of side QR ?

Correct Answer: 38 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

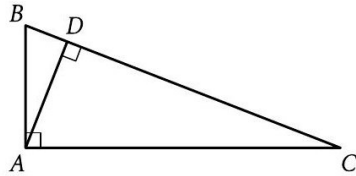
- Step 1: Let $QR = x$ (the length we want to find).
- Step 2: Since QR and PS are opposite sides of rectangle $PQRS$, they are equal: $PS = QR = x$.
- Step 3: We're told PS is 8 more than RS , so $RS = PS - 8 = x - 8$.
- Step 4: The diagonal QS connects Q and S , forming a right triangle with legs QR and RS : $QS^2 = QR^2 + RS^2$.
- Step 5: Substitute the known values: $2,344 = x^2 + (x-8)^2$.
- Step 6: Expand $(x-8)^2$: $x^2 - 16x + 64$. The equation becomes: $2,344 = x^2 + x^2 - 16x + 64$.
- Step 7: Combine like terms: $2,344 = 2x^2 - 16x + 64$.
- Step 8: Move everything to one side: $2x^2 - 16x + 64 - 2,344 = 0$, which simplifies to $2x^2 - 16x - 2,280 = 0$.
- Step 9: Divide the entire equation by 2: $x^2 - 8x - 1,140 = 0$.
- Step 10: Solve using the quadratic formula: $x = [8 \pm \sqrt{(64 + 4,560)}] / 2 = [8 \pm \sqrt{4,624}] / 2 = [8 \pm 68] / 2$.
- Step 11: Taking the positive root (since a length can't be negative): $x = (8+68)/2 = 76/2 = 38$.
- Step 12: This is a grid-in question — enter 38 as your final answer.

Right triangles and trigonometry — Q4 [Hard] [Grid-In] (ID: c4a9c94d)

Question ID: c4a9c94d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . If the length of $\overline{BD}$ is 5 and the tangent of angle CAD is 1.3, what is the length of $\overline{AD}$?

Correct Answer: 6.5 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Identify the key structure: angle $BAC = 90^\circ$ (right angle at A) and $AD \perp BC$ (right angle at D) means AD is the altitude from the right angle to the hypotenuse in right triangle ABC.
- Step 2: Recall the geometric mean relationship for this configuration: $AD^2 = BD \times DC$.
- Step 3: Use the given tangent value to find DC in terms of AD: in right triangle ACD (right angle at D), $\tan(\text{angle } CAD) = \text{opposite/adjacent} = DC/AD$. Given $\tan(CAD) = 1.3$, this means $DC = 1.3 \times AD$.
- Step 4: Substitute $BD = 5$ (given) and $DC = 1.3 \times AD$ into the geometric mean formula: $AD^2 = 5 \times (1.3 \times AD)$.
- Step 5: Simplify the right side: $AD^2 = 6.5 \times AD$.
- Step 6: Divide both sides by AD (valid since $AD \neq 0$): $AD = 6.5$.
- Step 7: This is a grid-in question — enter 6.5 as your final answer.

Right triangles and trigonometry — Q5 [Hard] (ID: f389569d)

Question ID: f389569d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

Answer

- A. 117
- B. $117\sqrt{3}$
- C. 234
- D. $1,521\sqrt{3}$

Correct Answer: C

Step-by-Step Solution:

Step 1: Recall the equilateral triangle height formula: height = side $\times (\sqrt{3}/2)$.

Step 2: Rearrange to solve for side: side = height $\div (\sqrt{3}/2)$ = height $\times (2/\sqrt{3})$.

Step 3: Substitute the given height: side = $39\sqrt{3} \times (2/\sqrt{3})$.

Step 4: Notice the $\sqrt{3}$ in the numerator and denominator cancel out: side = $39 \times 2 = 78$.

Step 5: Calculate the perimeter using the formula perimeter = $3 \times$ side: perimeter = $3 \times 78 = 234$.

Step 6: This matches choice C.

Right triangles and trigonometry — Q6 [Hard] (ID: e3158182)

Question ID: e3158182

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Acute angle P has a measure of p radians, and acute angle T has a measure of t radians. The function g is defined by $g(x) = p(8)^x + 6t$. If $\sin p = \cos t$, which of the following represents the value $g(0)$ in terms of t ?

Answer

- A. $6t + \pi$
- B. $6t$
- C. $5t + \pi$
- D. $5t + \frac{\pi}{2}$

Correct Answer: D

Step-by-Step Solution:

Step 1: Use the given relationship $\sin(p) = \cos(t)$ to establish that angles P and T are complementary: $p + t = \pi/2$ (since this cofunction identity holds specifically for complementary angles, measured in radians here since p and t are defined in radians).

Step 2: Solve this relationship for p : $p = \pi/2 - t$.

Step 3: Evaluate the function $g(x) = p(8)^x + 6t$ at $x = 0$: $g(0) = p(8)^0 + 6t$.

Step 4: Simplify using the zero exponent rule: $(8)^0 = 1$, so $g(0) = p(1) + 6t = p + 6t$.

Step 5: Substitute the expression for p from Step 2: $g(0) = (\pi/2 - t) + 6t$.

Step 6: Combine like terms: $-t + 6t = 5t$, so $g(0) = \pi/2 + 5t$, which can be written as $5t + \pi/2$.

Step 7: This matches choice D.

Right triangles and trigonometry — Q7 [Hard] [Grid-In] (ID: 98e02472)

Question ID: 98e02472

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle XYZ , angles X and Z are acute angles. Point R lies on XZ such that YR is perpendicular to XZ . The measure of $\angle XZY$ is a° , the measure of $\angle XYR$ is b° , and the length of XY is 19. If $\sin a^\circ = \frac{12}{13}$, what is the value of $\tan b^\circ$?

Correct Answer: 12/5 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Identify the right angle in triangle XYZ : since X and Z are both acute angles (given), the right angle must be at Y .
- Step 2: Since $YR \perp XZ$, this creates two smaller right triangles (XYR and YZR), each similar to the original triangle XYZ .
- Step 3: Identify the similarity: triangle XYR shares angle X with triangle XZY , and both have a right angle (at R and at Y respectively) — this makes them similar, with angle XYR corresponding to angle XZY .
- Step 4: Since angle $XYR = b^\circ$ and angle $XZY = a^\circ$, and these are corresponding angles in similar triangles, $b = a$.
- Step 5: Use the given $\sin(a^\circ) = \frac{12}{13}$ to find the full right triangle ratio: if the opposite side is 12 and the hypotenuse is 13, use the Pythagorean theorem to find the adjacent side: $\text{adjacent} = \sqrt{(13^2 - 12^2)} = \sqrt{(169 - 144)} = \sqrt{25} = 5$.
- Step 6: Calculate $\tan(a^\circ) = \text{opposite/adjacent} = \frac{12}{5}$.
- Step 7: Since $b = a$, $\tan(b^\circ) = \tan(a^\circ) = \frac{12}{5}$.
- Step 8: This is a grid-in question — enter 12/5 (or its decimal equivalent, 2.4) as your final answer.

Right triangles and trigonometry — Q8 [Hard] [Grid-In] (ID: 4fb972f2)

Question ID: 4fb972f2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle QRS , the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4. What is the length of side RS ?

Correct Answer: 36 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

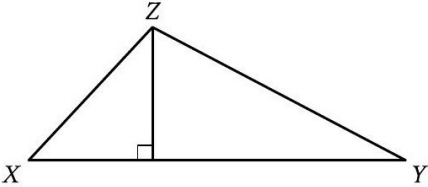
- Step 1: Identify the given information: hypotenuse $SQ = 4\sqrt{82}$, one leg $QR = 4$.
- Step 2: Set up the Pythagorean theorem to solve for the missing leg RS : $QR^2 + RS^2 = SQ^2$.
- Step 3: Rearrange to isolate RS^2 : $RS^2 = SQ^2 - QR^2$.
- Step 4: Calculate SQ^2 : $(4\sqrt{82})^2 = 4^2 \times (\sqrt{82})^2 = 16 \times 82 = 1,312$.
- Step 5: Calculate QR^2 : $4^2 = 16$.
- Step 6: Substitute these values: $RS^2 = 1,312 - 16 = 1,296$.
- Step 7: Take the square root of both sides: $RS = \sqrt{1,296} = 36$.
- Step 8: This is a grid-in question — enter 36 as your final answer.

Right triangles and trigonometry — Q9 [Hard] (ID: 6c67dd47)

Question ID: 6c67dd47

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 54° . The length of $\overline{XY}$ is 26 units and the length of $\overline{XZ}$ is 19 units. What is the area, in square units, of triangle XYZ ?

Answer

- A. 247
- B. 494
- C. $247 \sin 54^\circ$
- D. $494 \sin 54^\circ$

Correct Answer: C

Step-by-Step Solution:

- Step 1: Identify the altitude drawn from vertex Z down to side XY , creating a right angle at the foot of the altitude (let's call this point W).
- Step 2: In the right triangle formed (X, W, Z), the angle at X is given as 54° , and the hypotenuse of this small right triangle is $XZ = 19$.
- Step 3: Use the sine ratio to find the height ZW : $\sin(X) = \text{opposite/hypotenuse} = ZW/XZ$, so $ZW = XZ \times \sin(X) = 19 \times \sin(54^\circ)$.
- Step 4: Identify the base of the ORIGINAL triangle XYZ : this is side $XY = 26$.
- Step 5: Apply the triangle area formula: $\text{Area} = (1/2) \times \text{base} \times \text{height} = (1/2) \times 26 \times 19\sin(54^\circ)$.
- Step 6: Simplify: $(1/2) \times 26 = 13$, so $\text{Area} = 13 \times 19\sin(54^\circ) = 247\sin(54^\circ)$.
- Step 7: This matches choice C.

Circles

12 questions · Easy 1 / Medium 3 / Hard 8

Circles — Q1 [Easy] [Grid-In] (ID: 186fdbca)

Question ID: 186fdbca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Easy ☐ ☐ ☐

Question

A circle has a diameter of 12.6 centimeters. What is the radius of the circle, in centimeters?

Correct Answer: 6.3 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

Step 1: Recall the relationship between diameter and radius: radius = diameter $\div$ 2.

Step 2: Substitute the given diameter: radius = 12.6 $\div$ 2.

Step 3: Calculate: 12.6 $\div$ 2 = 6.3.

Step 4: This is a grid-in question — enter 6.3 as your final answer.

Circles — Q2 [Medium] (ID: 408b39ea)

Question ID: 408b39ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium ☐ ☐ ☐

Question

$$(x - 3)^2 + (y + 16)^2 = 289$$

The graph of the given equation is a circle in the xy -plane. The point (q, s) lies on the circle. Which of the following is NOT a possible value of q ?

Answer

- A. -17
- B. -3
- C. 3
- D. 17

Correct Answer: A

Step-by-Step Solution:

Step 1: Identify the center and radius from the equation $(x-3)^2 + (y+16)^2 = 289$: center = $(3, -16)$, radius = $\sqrt{289} = 17$.

Step 2: Determine the range of possible x -coordinates for any point on the circle: from (center x - radius) to (center x + radius), which is $(3-17)$ to $(3+17)$, or -14 to 20 .

Step 3: Check each answer choice against this range: -17 is LESS than -14 , meaning it falls outside the circle's possible x -range.

Step 4: Since -17 cannot be a valid x -coordinate for any point on this circle, it is NOT a possible value of q .

Step 5: This matches choice A.

Circles — Q3 [Medium] (ID: 267f82b5)

Question ID: 267f82b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

A circle in the xy -plane has the equation $(x - 19)^2 + (y - 17)^2 = 64k^2$, where k is a positive constant. Which of the following gives the center of the circle and its radius?

Answer

- A. The center is at $(19, 17)$, and the radius is 8.
- B. The center is at $(19, 17)$, and the radius is $8k$.
- C. The center is at $(19, 17)$, and the radius is 64.
- D. The center is at $(19, 17)$, and the radius is $64k$.

Correct Answer: B

Step-by-Step Solution:

Step 1: Identify the center directly from the equation's structure $(x-19)^2 + (y-17)^2$: the center is $(19, 17)$.

Step 2: Identify the radius by taking the square root of the right-hand side: radius $= \sqrt{64k^2}$.

Step 3: Simplify this square root by taking the square root of each factor separately: $\sqrt{64} \times \sqrt{(k^2)} = 8 \times k$ (since k is given as positive, $\sqrt{(k^2)}$ simplifies directly to k , no absolute value needed).

Step 4: The radius is $8k$.

Step 5: This matches choice B: center $(19,17)$, radius $8k$.

Circles — Q4 [Medium] (ID: 11301fb6)

Question ID: 11301fb6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

The difference between the measure of angle A and the measure of angle B is $-\frac{5}{6}\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B , in degrees?

Answer

- A. $-\frac{5}{6}\pi \cdot \frac{360^\circ}{\pi}$
- B. $-\frac{5}{6}\pi \cdot \frac{180^\circ}{\pi}$
- C. $-\frac{5}{6}\pi \cdot \frac{\pi}{180^\circ}$
- D. $-\frac{5}{6} \cdot \frac{180^\circ}{\pi}$

Correct Answer: C

Step-by-Step Solution:

Step 1: Recall the radians-to-degrees conversion factor: multiply the radian measure by $(180^\circ/\pi)$.

Step 2: Apply this to the given difference, $-(5/6)\pi$ radians: multiply by $(180^\circ/\pi)$.

Step 3: Set up the expression: $-(5/6)\pi \times (180^\circ/\pi)$.

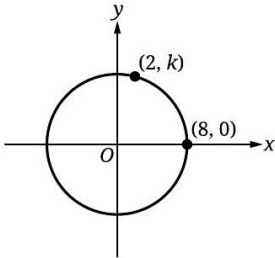
Step 4: This exact structure — the original radian expression multiplied by $(180^\circ/\pi)$ — matches choice C.

Circles — Q5 [Hard] (ID: d12bca0a)

Question ID: d12bca0a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Note: Figure not drawn to scale.

The circle shown has its center at $(0,0)$. What is the value of k ?

Answer

- A. 6
- B. 7
- C. 8
- D. $\sqrt{60}$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Use the given point $(8,0)$ to find the radius: since this point is on the x-axis and the circle is centered at the origin, the distance from the origin to $(8,0)$ IS the radius: $r = 8$.
- Step 2: Write the circle's equation using this radius: $x^2 + y^2 = 8^2 = 64$.
- Step 3: Substitute the other given point $(2,k)$ into this equation: $2^2 + k^2 = 64$.
- Step 4: Simplify: $4 + k^2 = 64$.
- Step 5: Solve for k^2 : $k^2 = 64 - 4 = 60$.
- Step 6: Take the square root: $k = \sqrt{60}$ (taking the positive root, since the point appears above the x-axis in the figure).
- Step 7: This matches choice D.

Circles — Q6 [Hard] (ID: 8493fd10)

Question ID: 8493fd10

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$$

The given equation, where t is a constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{83}$. What is the value of t ?

Answer

- A. 73
- B. 71
- C. 37
- D. 35

Correct Answer: B

Step-by-Step Solution:

- Step 1: Group the x -terms and y -terms separately: $(x^2 + x\sqrt{t}) + (y^2 - y) - 65 = 0$.
- Step 2: Complete the square for the x -terms: take half of the coefficient of x (which is $\sqrt{t}$), giving $\sqrt{t}/2$; square this to get $t/4$. Rewrite: $x^2 + x\sqrt{t} = (x + \sqrt{t}/2)^2 - t/4$.
- Step 3: Complete the square for the y -terms: take half of the coefficient of y (which is -1), giving $-1/2$; square this to get $1/4$. Rewrite: $y^2 - y = (y - 1/2)^2 - 1/4$.
- Step 4: Substitute these completed-square forms back into the original equation: $(x + \sqrt{t}/2)^2 - t/4 + (y - 1/2)^2 - 1/4 - 65 = 0$.
- Step 5: Move all the constant terms to the right side: $(x + \sqrt{t}/2)^2 + (y - 1/2)^2 = 65 + 1/4 + t/4 = 65.25 + t/4$.
- Step 6: Since the radius is given as $\sqrt{83}$, the right side of the standard-form equation must equal 83 (radius squared): $65.25 + t/4 = 83$.
- Step 7: Solve for t : $t/4 = 83 - 65.25 = 17.75$.
- Step 8: Multiply both sides by 4: $t = 17.75 \times 4 = 71$.
- Step 9: This matches choice B.

Circles — Q7 [Hard] (ID: a5ff1a96)

Question ID: a5ff1a96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard ■■■■■

Question

In the xy -plane, circle M is the graph of the equation $(x - 4)^2 + (y - 5)^2 = 9$. Circle P has the same center as circle M but has a radius that is twice the radius of circle M . Which equation represents circle P ?

Answer

- A. $(x - 4)^2 + (y - 5)^2 = 18$
- B. $(x - 4)^2 + (y - 5)^2 = 36$
- C. $(x - 8)^2 + (y - 10)^2 = 9$
- D. $(x - 8)^2 + (y - 10)^2 = 18$

Correct Answer: B

Step-by-Step Solution:

- Step 1: Identify circle M 's center and radius from its equation $(x-4)^2 + (y-5)^2 = 9$: center = $(4,5)$, radius = $\sqrt{9} = 3$.
- Step 2: Since circle P has the SAME center as circle M , its equation will also have $(x-4)^2$ and $(y-5)^2$ unchanged.
- Step 3: Calculate circle P 's radius: twice circle M 's radius = $2 \times 3 = 6$.
- Step 4: Calculate the new right-hand side (radius squared) for circle P 's equation: $6^2 = 36$.
- Step 5: Write circle P 's full equation: $(x-4)^2 + (y-5)^2 = 36$.
- Step 6: This matches choice B.

Circles — Q8 [Hard] [Grid-In] (ID: 149021da)

Question ID: 149021da

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard ■■■■■

Question

A circle has center P , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 4π units. What is the length, in units, of the radius of the circle?

Correct Answer: 16 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Recall the arc length formula: arc length = $(\text{central angle} / 360^\circ) \times 2\pi r$.
- Step 2: Substitute the given values: $4\pi = (45/360) \times 2\pi r$.
- Step 3: Simplify the fraction $45/360$ by dividing both by 45: $45/360 = 1/8$.
- Step 4: The equation becomes: $4\pi = (1/8) \times 2\pi r = (2\pi r)/8 = \pi r/4$.
- Step 5: Multiply both sides by 4 to clear the fraction: $16\pi = \pi r$.
- Step 6: Divide both sides by π : $r = 16$.
- Step 7: This is a grid-in question — enter 16 as your final answer.

Circles — Q9 [Hard] (ID: 10e059a6)

Question ID: 10e059a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard ■■■■■

Question

The measure of angle A is $\frac{\pi}{216}$ radians. The measure of angle B is 36 times the measure of angle A . What is the value of $\cos B$?

Answer

- A. 0
- B. $\frac{1}{2}$
- C. $\frac{\sqrt{2}}{2}$
- D. $\frac{\sqrt{3}}{2}$

Correct Answer: D

Step-by-Step Solution:

- Step 1: Identify the measure of angle A : $\pi/216$ radians (given).
- Step 2: Calculate angle B using the given relationship: angle $B = 36 \times$ angle $A = 36 \times (\pi/216)$.
- Step 3: Simplify this multiplication: $36 \times \pi/216 = 36\pi/216$.
- Step 4: Reduce the fraction $36/216$ by dividing both by 36: $36/216 = 1/6$.
- Step 5: So angle $B = \pi/6$ radians.
- Step 6: Recall the standard value: $\cos(\pi/6) = \sqrt{3}/2$ (this is one of the standard reference angles, equivalent to 30°).
- Step 7: This matches choice D.

Circles — Q10 [Hard] [Grid-In] (ID: 9f9112ab)

Question ID: 9f9112ab

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard ■■■■■

Question

The measure of angle L is 60 degrees. If the measure of angle L is $\frac{\pi}{n}$ radians, where n is a constant, what is the value of n ?

Correct Answer: 3 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

- Step 1: Convert the given angle measure from degrees to radians: $60^\circ \times (\pi/180^\circ)$.
- Step 2: Simplify the fraction $60/180$ by dividing both by 60: $60/180 = 1/3$.
- Step 3: So $60^\circ = \pi/3$ radians.
- Step 4: Compare this to the given form π/n : since angle $L = \pi/3 = \pi/n$, this means $n = 3$.
- Step 5: This is a grid-in question — enter 3 as your final answer.

Circles — Q11 [Hard] [Grid-In] (ID: 27f1db42)

Question ID: 27f1db42

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, circle F is defined by the equation $(x + 8)^2 + (y + 8)^2 = 25$. Circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Circle G is defined by the equation $(x + 8)^2 + (y + 8)^2 = p$, where p is a constant. What is the value of p ?

Correct Answer: 49 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

Step 1: Identify circle F 's radius from its equation $(x+8)^2 + (y+8)^2 = 25$: $\text{radius}^2 = 25$, so $\text{radius} = \sqrt{25} = 5$.

Step 2: Apply the given relationship: circle G 's radius is 2 units GREATER than circle F 's radius (an ADDITION, not a multiplication): circle G 's radius $= 5 + 2 = 7$.

Step 3: Since circle G 's equation is written as $(x+8)^2 + (y+8)^2 = p$, the value p represents circle G 's radius SQUARED.

Step 4: Calculate p : $p = 7^2 = 49$.

Step 5: This is a grid-in question — enter 49 as your final answer.

Circles — Q12 [Hard] [Grid-In] (ID: 44d67c6c)

Question ID: 44d67c6c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + y^2 - 10x - 8y - 14n = 0$$

The given equation represents circle A in the xy -plane, where n is a constant. Point $(11, 8)$ lies on circle B , which has the same center but twice the diameter as circle A . What is the value of n ?

Correct Answer: -2 (Student-Produced Response — no answer choices given)

Step-by-Step Solution:

Step 1: Complete the square for the x -terms in $x^2 - 10x$: take half of -10 (giving -5), square it (giving 25), rewrite as $(x-5)^2 - 25$.

Step 2: Complete the square for the y -terms in $y^2 - 8y$: take half of -8 (giving -4), square it (giving 16), rewrite as $(y-4)^2 - 16$.

Step 3: Substitute back into the original equation: $(x-5)^2 - 25 + (y-4)^2 - 16 - 14n = 0$.

Step 4: Rearrange to standard form: $(x-5)^2 + (y-4)^2 = 25 + 16 + 14n = 41 + 14n$. This means circle A has center $(5,4)$ and $\text{radius}^2 = 41+14n$.

Step 5: Circle B shares the same center $(5,4)$ but has TWICE the diameter of circle A — meaning circle B 's radius is also twice circle A 's radius.

Step 6: Use the given point $(11,8)$ on circle B to find circle B 's actual radius via the distance formula from the center $(5,4)$: $\text{distance} = \sqrt{[(11-5)^2 + (8-4)^2]} = \sqrt{(36+16)} = \sqrt{52}$. So circle B 's $\text{radius}^2 = 52$.

Step 7: Since circle B 's radius $= 2 \times$ circle A 's radius, squaring both sides: circle B 's $\text{radius}^2 = 4 \times$ circle A 's radius^2 . Substitute: $52 = 4 \times (41+14n)$.

Step 8: Divide both sides by 4: $13 = 41 + 14n$.

Step 9: Solve for n : $14n = 13 - 41 = -28$, so $n = -28/14 = -2$.

Step 10: This is a grid-in question — enter -2 as your final answer.

June Resit 2026 - Math (Part 1)

Εκφωνήσεις & αναλυτικές λύσεις

SAT Math - Σημειώσεις

Άσκηση 1 (Exponential - ισοδύναμες μορφές)

The function $m(x) = c(1.8^x + 1.8^d)$, $c, d \in \mathbb{Z}$, $0 < c < d$. The functions $s(x), t(x)$ are given by

$$\text{I. } s(x) = c \cdot (1.8^x + u) \quad \text{II. } t(x) = c \cdot 1.8^x + v$$

Which of the two has the same y -intercept as $m(x)$?

A) I B) II C) I & II D) Neither

Λύση

Το y -intercept κάθε συνάρτησης είναι η τιμή της στο $x = 0$, και $1.8^0 = 1$:

$$m(0) = c(1 + 1.8^d) = c + c \cdot 1.8^d$$

$$s(0) = c(1 + u) = c + cu$$

$$t(0) = c \cdot 1 + v = c + v$$

Για να ισχύει $s(0) = m(0)$ πρέπει $u = 1.8^d$, και για να ισχύει $t(0) = m(0)$ πρέπει $v = c \cdot 1.8^d$. Οι u, v είναι τυχαίες σταθερές, άρα καμία από τις δύο ισότητες δεν ισχύει αναγκαστικά.

Answer / Απάντηση: D) Neither

Σημείωση: Αν διαβάσουμε την εκφώνηση ότι οι s, t είναι ταυτοτικά ίσες με την m , τότε $u = 1.8^d$, $v = c \cdot 1.8^d$ και θα είχαν το ίδιο y -intercept (C). Η λύση του αρχείου εννοεί u, v ελεύθερες σταθερές, γι' αυτό η απάντηση είναι D.

Άσκηση 2 (Vieta)

$120x^2 - (15p + 21q)x + pq = 0$, $p, q \in \mathbb{R}_{>0}$. The sum of solutions of the quadratic is in the form $k \cdot (5p + 7q)$. Find k .

Λύση

Για $ax^2 + bx + c = 0$ το άθροισμα των ριζών είναι $S = -\frac{b}{a}$ (τύποι Vieta). Εδώ $a = 120$, $b = -(15p + 21q)$:

$$S = -\frac{-(15p + 21q)}{120} = \frac{15p + 21q}{120} = \frac{3(5p + 7q)}{120} = \frac{1}{40}(5p + 7q).$$

Συγκρίνοντας με τη μορφή $k(5p + 7q)$ παίρνουμε

Answer / Απάντηση: $k = \frac{1}{40}$

Άσκηση 3 (Εκθετική - y -τομή)

$f(x) = -a^x + b$. Let $g(x) = f(x) - 9$, which has y -intercept at $(0, -\frac{31}{4})$. If $a \cdot b = \frac{225}{4}$, find a and b .

Λύση

Αφού $a^0 = 1$:

$$g(0) = f(0) - 9 = (-1 + b) - 9 = b - 10.$$

Η y -τομή είναι $-\frac{31}{4}$, άρα

$$b - 10 = -\frac{31}{4} \Rightarrow b = 10 - \frac{31}{4} = \frac{9}{4}.$$

Από το γινόμενο:

$$a = \frac{225/4}{b} = \frac{225/4}{9/4} = 25.$$

Έλεγχος: $ab = 25 \cdot \frac{9}{4} = \frac{225}{4} \checkmark$ και $g(0) = \frac{9}{4} - 10 = -\frac{31}{4} \checkmark$

Answer / Απάντηση: $a = 25, b = \frac{9}{4}$

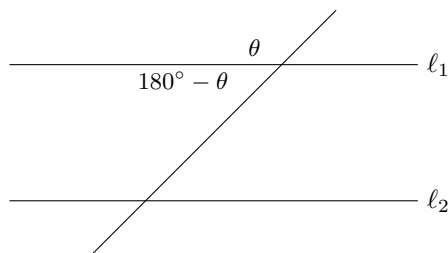
Σημείωση: στη λύση του αρχείου είχε παραλειφθεί το -9 ($g(0) = 1 + b$), που έδινε λάθος τιμές. Εδώ η g υπολογίζεται σωστά.

Άσκηση 4 (Παράλληλες & τέμνουσα)

A transversal line intersects two parallel lines, creating 4 acute and 4 obtuse angles. The measure of one of these angles is $(9x - 150)$ degrees. The sum of four of these angles is k degrees. Which of the following is **NOT** a possible value for k ?

(A) $18x - 120$ (B) $-18x + 1140$ (C) 360 (D) $-36x + 1320$

Λύση



Όταν μια τέμνουσα κόβει δύο παράλληλες, και οι οκτώ γωνίες παίρνουν μόνο δύο τιμές: θ (4 γωνίες) και $180^\circ - \theta$ (4 γωνίες), με $\theta = 9x - 150$. Άρα $180^\circ - \theta = 330 - 9x$.

Αθροίζοντας τέσσερις γωνίες, έστω j από αυτές ίσες με θ και $4 - j$ ίσες με $180^\circ - \theta$:

j	Άθροισμα k
4	$4\theta = 36x - 600$
3	$3\theta + (180 - \theta) = 2\theta + 180 = 18x - 120$
2	$2\theta + 2(180 - \theta) = 360$
1	$\theta + 3(180 - \theta) = 540 - 2\theta = 840 - 18x$
0	$4(180 - \theta) = 1320 - 36x$

Οι επιλογές: (A) $18x - 120 \checkmark$ (εμφανίζεται), (C) $360 \checkmark$, (D) $1320 - 36x \checkmark$. Η (B) $-18x + 1140$ δεν εμφανίζεται (η σωστή τιμή είναι $-18x + 840$).

Answer / Απάντηση: B

Άσκηση 5 (Τετραγωνική - τομές με x -άξονα)

The quadratic $g(x) = ax^2 + bx + c$ crosses the x -axis at $(x_1, 0), (x_2, 0)$ where $x_1 + x_2 = -6$, $x_1x_2 = -27$. If $a > 1$ and $a \in \mathbb{Z}$, what could be the value of $b - a$?

(A) 10 (B) 5 (C) 8 (D) 12

Λύση

Βήμα 1 - οι ρίζες. Οι x_1, x_2 είναι ρίζες της $t^2 - (x_1 + x_2)t + x_1x_2 = t^2 + 6t - 27 = (t + 9)(t - 3)$, άρα $x_1, x_2 = -9, 3$ (με Desmos: τομή των $y = -6 - x$ και $y = -27/x$).

Βήμα 2 - η μορφή της g .

$$g(x) = a(x + 9)(x - 3) = a(x^2 + 6x - 27) = ax^2 + 6ax - 27a.$$

Άρα $b = 6a$, και

$$b - a = 5a.$$

Βήμα 3 - περιορισμός. Αφού $a \in \mathbb{Z}$ και $a > 1$, ισχύει $a \geq 2$, άρα $b - a \in \{10, 15, 20, \dots\}$: πολλαπλάσιο του 5 και τουλάχιστον 10.

- (A) $10 = 5 \cdot 2$ ✓ (B) $5 \Rightarrow a = 1 \times$ (θέλουμε $a > 1$)
- (C) 8, (D) 12: δεν είναι πολλαπλάσια του 5 ×

Answer / Απάντηση: A) 10

Άσκηση 6 (Συνάρτηση τετραγωνικής ρίζας)

$g(x) = k\sqrt{x + m}$, the point $(-49, 0)$ lies on the curve and $g(32) < 0$. Which of the following must be true about g ?

(A) $k < m$ (B) $k > m$ (C) $g(0) = 49$ (D) $g(0) = -49$

Λύση

Από το σημείο $(-49, 0)$: $g(-49) = k\sqrt{m - 49} = 0$, άρα $k = 0$ ή $m = 49$.

Αν $k = 0$ τότε $g \equiv 0$ και $g(32) = 0$, ενώ δίνεται $g(32) < 0$. Άρα $m = 49$ (και $k \neq 0$).

Τώρα $g(32) = k\sqrt{32 + 49} = k\sqrt{81} = 9k < 0 \Rightarrow k < 0$.

Έτσι $k < 0 < 49 = m$, δηλαδή $k < m$. Για τις υπόλοιπες: $g(0) = k\sqrt{49} = 7k$, που είναι αρνητικός αλλά όχι απαραίτητα -49 (ούτε 49), άρα οι (C), (D) δεν ισχύουν αναγκαστικά.

Answer / Απάντηση: A) $k < m$

Άσκηση 7 (Παραγοντοποίηση)

$x(9x - 2k) + m(9x - 2k) - (x + m) = 0$, with k, m positive constants. The equation has a solution at $x = 50$. Find k .

Λύση

Βγάζουμε κοινό παράγοντα το $(9x - 2k)$ από τους δύο πρώτους όρους:

$$(x + m)(9x - 2k) - (x + m) = 0 \Rightarrow (x + m)(9x - 2k - 1) = 0.$$

Άρα $x = -m$ ή $x = \frac{2k + 1}{9}$. Αφού $m > 0$, το $-m$ είναι αρνητικό και δεν μπορεί να είναι 50. Επομένως

$$\frac{2k + 1}{9} = 50 \Rightarrow 2k + 1 = 450 \Rightarrow k = \frac{449}{2} = 224.5.$$

Έλεγχος (ανεξαρτησία από το m): $9 \cdot 50 - 2(224.5) - 1 = 450 - 449 - 1 = 0$ ✓

Answer / Απάντηση: $k = 224.5$

Άσκηση 8 (Κυβική - πρόσημο)

$h(x) = -5(x-u)(x-v)(x-w)$. For $x \in (u, w)$ (that is $u < x < w$) $h(x)$ is negative. The graph of $y = h(x)$ contains the point (m, n) , $m, n \in \mathbb{R}$. If $n = -12$, which of the following is/are true?

- (I) $m < w$ (II) $u < m < v$ (III) $m > u$
 (A) Only I (B) Only II (C) Only I, II (D) Only I, III

Λύση

Για $x \in (u, w)$ ισχύει $x - u > 0$ και $x - w < 0$, άρα $(x - u)(x - w) < 0$ και $-5(x - u)(x - w) > 0$. Για να είναι $h(x) < 0$ πρέπει ο τρίτος παράγοντας $x - v$ να είναι αρνητικός, δηλαδή $x < v$, για κάθε x του (u, w) . Άρα $v \geq w$, και τα πρόσημα της h είναι:

$x < u$	$u < x < w$	$w < x < v$	$x > v$
+	-	+	-

Αφού $h(m) = -12 < 0$, το m ανήκει σε περιοχή αρνητικού προσήμου, δηλαδή $m \in (u, w) \cup (v, \infty)$. Και στις δύο περιπτώσεις $m > u$, άρα η **(III) ισχύει πάντα**. Οι (I) και (II) ισχύουν μόνο όταν το m είναι στο (u, w) , ενώ αν $m > v$ δεν ισχύουν.

Answer / Απάντηση: D) Only I, III (κλειδί του αρχείου)

Προσοχή: με την εκφώνηση όπως έχει μεταφερθεί, η μόνη πάντα αληθής πρόταση είναι η (III) και καμία επιλογή δεν είναι ακριβώς "Only (III)". Η D είναι η πλησιέστερη (και το κλειδί της εξέτασης), αλλά αξίζει έλεγχος της πρωτότυπης εκφώνησης.

Άσκηση 9 (Εκθετική)

$f(x) = c \cdot d^x$, $c, d > 0$. If $f(r) = k$ and $f(r+1) = k - 0.64k$, find d .

Λύση

Η νέα τιμή είναι $f(r+1) = k - 0.64k = 0.36k$. Ο λόγος διαδοχικών τιμών της εκθετικής είναι σταθερός και ίσος με τη βάση:

$$\frac{f(r+1)}{f(r)} = \frac{cd^{r+1}}{cd^r} = d.$$

Άρα

$$d = \frac{0.36k}{k} = 0.36.$$

(Μείωση 64% ανά βήμα σημαίνει πολλαπλασιασμός με $1 - 0.64 = 0.36$.)

Answer / Απάντηση: $d = 0.36$

Άσκηση 10 (Ρίζες τετραγωνικών)

$f(x) = x^2 + 3x - 10$ has solutions $r < s$; $g(x) = x^2 + 4x - 21$ has solutions $u < v$. $h(x) = x^2 + 7x + c$ has solutions $s + u$ and $v + r$ and can be rewritten as $(x + a)(x + b)$. Find c .

Λύση

Ρίζες της f : $x^2 + 3x - 10 = (x + 5)(x - 2) = 0 \Rightarrow r = -5, s = 2$.

Ρίζες της g : $x^2 + 4x - 21 = (x + 7)(x - 3) = 0 \Rightarrow u = -7, v = 3$.

Ρίζες της h :

$$s + u = 2 - 7 = -5, \quad v + r = 3 - 5 = -2.$$

Άρα $h(x) = (x + 5)(x + 2) = x^2 + 7x + 10$. **Έλεγχος:** άθροισμα ριζών $-5 + (-2) = -7 = -(7) \checkmark$

Answer / Απάντηση: $c = 10$